Unified Rare-Variant Meta-Regression Integrating Deep-Learning Structural Pathogenicity and Base-Pair Population Constraint Uncovers Epithelial Hemidesmosome, Xenophagy, and Immunometabolic Architectures in Inflammatory Bowel Disease

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Abstract

Background: Gene-based rare-variant association studies (RVAS) in Inflammatory Bowel Disease (IBD) traditionally partition coding variants into discrete, mutually exclusive filtering masks (e.g., high-confidence protein-truncating variants [`pLoF`] or deleterious missense thresholds), discarding quantitative structural pathogenicity gradients and suffering severe power loss when pathogenic variants span both truncating and missense classes. Recently, Aguilar, Rivas & Rivas (2025) introduced a unified weighted least-squares (WLS) rare-variant meta-regression framework that regresses signed variant-level log-odds ratios ($\hat{\beta}_i$) directly onto continuous functional annotations within each protein-coding gene. However, standard formulations that simultaneously include an intercept (**1**), a binary `pLoF` indicator (pLoF_ind), and a binary missense indicator (missense_ind) in non-synonymous coding regressions induce exact or near-exact design-matrix rank deficiency ($\mathbf{1} = \mathbf{x}_{\mathrm{LoF}} + \mathbf{x}_{\mathrm{Mis}}$), collapsing the partial R^2 of protein-truncating variants in 88.1% of human genes and inflating variance inflation factors (VIF > 50) in the remaining 11.9% of genes containing initiation codon (`start_lost`) alleles.

Methods: We developed and executed a collinearity-free unified WLS rare-variant meta-regression across the IBD Exome Sequencing Results Browser dataset (`IBD`, Crohn's Disease [`CD`], and Ulcerative Colitis [`UC`]; 16,820,350 merged autosomal coding variant records across 17,974 protein-coding genes) at allele frequency thresholds AF ≤ 0.001 and AF ≤ 0.0001. For the 59.68% of rare coding variants at AF ≤ 0.001 (64.12% at AF ≤ 0.0001, including 100% of singletons) lacking browser meta-analysis effect sizes (BETA_meta) or standard errors, we derived exact signed log-odds ratios and asymptotic variances from raw case–control allelic counts using the Firth (1993) Jeffreys-prior penalized 2×2 likelihood estimator. We eliminated both evolutionary GERP scores (`GERP_RS`) and the collinear binary indicator `missense_ind`, establishing missense/non-LoF coding alleles as the clean reference baseline (pLoF_ind $= 0$) and evaluating all seven hierarchical combinations (**M1**–**M7**) of three orthogonal moderators: continuous AlphaMissense pathogenicity ($\log \text{pathogenicity}_i = -\log(1 - s_{\mathrm{AM},i})$), binary protein truncation (pLoF_ind$_i$), and base-pair WES hidden Markov model mutational constraint probabilities from Aguilar, Rivas & Rivas ($\log \text{constraint}_i = -\log(1 - p_{\mathrm{con},i})$). We further characterized the analytical and empirical scaling between unified model significance ($-\log_{10} p_{\mathrm{M7}}$), explained variance ($R^2$), and canonical protein length in amino acids (L_{AA}), and performed multi-database pathway enrichment (GO, Reactome, KEGG, WikiPathways) and systems biology dissection.

Results: Removing `missense_ind` reduced the median maximum VIF across the three moderators in **M7** to **1.022** (AF ≤ 0.001, non-synonymous) and **1.014** (synonymous-referenced), restoring true orthogonal partial R^2 decomposition across the exome. Across 51,484 non-synonymous gene–phenotype tests at AF ≤ 0.001, exome-wide significant associations ($p < 3.4 \times 10^{-7}$) increased more than 6-fold from **37** in **M1** (`LoF_only`) and **170** in **M2** (`AM_only`) to **241** pairs (189 unique genes) in **M5** (`AM + LoF`) and **208** pairs (162 unique genes; 208 unique genes across M5 $\cup$ M7) in **M7** (`AM + LoF + Constraint`), with pLoF_ind independently significant ($p < 0.05$) in 69.2% of discovered loci alongside AlphaMissense (93.8%) and HMM constraint (30.3%). At the ultra-rare threshold AF ≤ 0.0001, discoveries progressed from **3** (**M1**) to **18** pairs (16 unique genes) in **M5** and **9** in **M7** (*NOD2*, *SACS*, *TTN*, *LAMA5*, *PLEC*, *ZNF462*, *HPD*, *DDX24*, *BSN*, *PFKFB3*, *RUNX1*, *DPP9*, *MYLK2*). Under the synonymous-referenced specification at AF ≤ 0.001, discoveries grew monotonically across moderator complexity: **39** (**M1**) $\rightarrow$ **189** (**M2**) $\rightarrow$ **255** (**M5**) $\rightarrow$ **266** (**M7**, 207 unique genes). Analysis of canonical protein length (L_{AA}) revealed a strong linear relationship between L_{AA} and rare coding variant count M_g ($r = +0.905$, ≈ 0.34 variants/AA), yielding a positive correlation between $\log_{10}(L_{\mathrm{AA}})$ and $-\log_{10}(p_{\mathrm{M7}})$ ($r = +0.321$, Spearman $\rho = +0.274$) governed by the inverse critical variance law $R^2_{\mathrm{crit}}(L_{\mathrm{AA}}) \approx 31.8/(0.34 L_{\mathrm{AA}} + 27.8)$: compact immune and metabolic sensors ($L_{\mathrm{AA}} < 500$ AA, e.g., *HPD*, *AGER*, *RUNX1*) require large explained variance ($R^2 \approx$ 19–28%) to reach exome-wide significance, whereas giant multi-domain structural scaffolds ($L_{\mathrm{AA}} > 2{,}000$ AA, e.g., *LAMA5*, *PLEC*, *SYNE1*) cross

significance at $R^2 \approx 1.8$–3.8%. Unbiased pathway enrichment of the 208 significant genes revealed nine cohesive biological systems: (1) the intestinal epithelial **Type I hemidesmosome and lamina densa basement membrane complex** (*PLEC*, *DST*, *ITGB4*, *LAMA5*, *COL7A1*, *HSPG2*, *HMCN1*, *TLN2*; OR = 96.5, $p = 2.17 \times 10^{-5}$, predominantly in UC), (2) the **apical tight/adherens junction and MLCK actomyosin contractility ring** (*CLDN20*, *AFDN*, *CGNL1*, *MYLK*, *MYLK2*, *CRACD*, *ADD3*, *SYNE1/2*), (3) **Rho/Rac/Cdc42 GTPase exchange factors** (*TIAM1*, *TRIO*, *DOCK3/7/9*, *PLEKHG4*, *IQSEC1*; $p = 3.29 \times 10^{-5}$), (4) **innate bacterial sensing, DPP9–NLRP1/CARD8 inflammasome restraint, and lymphocyte CBM/STAT3 signaling** (*NOD2*, *SBNO2*, *DPP9*, *CARD11*, *ZNF341*, *PRKCD*, *RUNX1*, *AGER*, *RNF213*, *SMURF1*), (5) **sequential antibacterial xenophagy, HOPS/CORVET tethering, and Paneth/goblet cell ERAD** (*PIK3R4*, *WIPI2* [direct ATG16L1 recruiter], *ATG2A*, *VPS13D/B*, *PIKFYVE*, *VPS11*, *PLEKHM2*, *SEL1L*), (6) **colonic mucin gel O-glycosylation and bile transport** (*MUC5B*, *MUC16*, *B4GALNT4*, *ABCB11*), (7) **epithelial Ca^{2+} signaling and goblet cell exocytosis channels** (*ITPR1*, *ATP2A3*, *TRPM7*, *CACNA1G*; $p = 4.49 \times 10^{-5}$), (8) **SWI/SNF and H3K4/H3K36 chromatin remodeling** (*SMARCA2*, *KMT2D*, *ASH1L*, *ZNF462*), and (9) **gut microbial acetate-to-nuclear acetyl-CoA immunometabolism** (*ACSS2*, *ACLY*, *PFKFB3*, *HPD*).

Conclusions: Eliminating binary missense indicator collinearity and integrating Firth penalized-likelihood rare-variant estimation into unified WLS meta-regression substantially increases gene discovery power in IBD, revealing that Ulcerative Colitis is heavily driven by rare coding perturbations in basal hemidesmosome–laminin anchoring, mucin O-glycosylation, and goblet cell ERAD, whereas Crohn's Disease centers on intracellular bacterial sensing, WIPI2–ATG16L1 xenophagy, and perijunctional MLCK actomyosin contractility.

Keywords: Inflammatory Bowel Disease · Crohn's Disease · Ulcerative Colitis · Whole-Exome Sequencing · Rare-Variant Meta-Regression · AlphaMissense · Genomic Constraint · Firth Penalized Likelihood · Hemidesmosome · Autophagy · Gene Size Scaling

1 Introduction

Inflammatory Bowel Disease (IBD), comprising Crohn's Disease (CD) and Ulcerative Colitis (UC), is a chronic, relapsing-remitting immune-mediated disorder of the gastrointestinal tract arising from a complex interplay between host genetic susceptibility, the luminal gut microbiota, and mucosal barrier integrity [6, 7]. While genome-wide association studies (GWAS) of common variants have identified more than 240 susceptibility loci [7], pinpointing the causal effector genes and translating non-coding signals into actionable cellular pathways remain major challenges. Whole-exome sequencing (WES) of large case–control cohorts directly interrogates protein-altering alleles across the allele frequency spectrum, offering a direct link between coding variation, protein structure, and disease mechanism [5].

Standard gene-based rare-variant association studies (RVAS) rely on burden tests or variance-component tests (such as SKAT-O) applied across a combinatorial grid of discrete variant filtering masks (e.g., high-confidence protein-truncating variants [`pLoF`] vs. all missense variants vs. missense variants exceeding an arbitrary pathogenicity threshold across multiple allele frequency cutoffs) [5, 8]. Despite notable successes—such as confirming *NOD2*, *IL23R*, and *CARD9* in Crohn's disease—discrete mask-based testing suffers from three fundamental limitations:

1. **Threshold Discretization and Loss of Quantitative Structural Gradients**: Treating all missense variants within a mask as exchangeable discards the continuous biophysical gradient of amino acid substitutions across protein domains, catalytic pockets, and protein–protein interfaces captured by deep-learning structural models such as AlphaMissense [3] and base-pair HMM population mutational constraint probabilities [1].
2. **Directional Cancellation Between Gain- and Loss-of-Function Alleles**: Standard burden tests sum allele counts across a gene without regard to whether pathogenic variants increase risk ($\hat{\beta}_i > 0$, e.g., loss of mucosal barrier function in *LAMA5* or *NOD2*) or confer protection ($\hat{\beta}_i < 0$, e.g., loss of pro-inflammatory signaling in *IL23R* or *TYK2*).
3. **Exclusion of Ultra-Rare Singletons Lacking Asymptotic Score-Test Variances**: In public exome sequencing browsers, nearly 60% of rare coding variants at AF ≤ 0.001 (and $> 64\%$ at AF ≤ 0.0001, including 100% of singletons with MAC $= 1$) have missing meta-analysis effect sizes (`BETA_meta = NA`) because standard inverse-variance meta-analysis drops strata with zero counts in cases or controls.

To overcome these barriers, Aguilar, Rivas & Rivas (2025) [1] and Lauer & Rivas (2025) [2] developed a unified weighted least-squares (WLS) rare-variant meta-regression framework (`wgs-constraint-llm`). Rather than collapsing variants into binary masks, the unified model treats each observed coding variant $i \in \{1, \ldots, M_g\}$ within gene g as a regression observation, modeling its signed log-odds ratio $\hat{\beta}_i$ (weighted by inverse variance $w_i = 1/\widehat{\mathrm{SE}}_i^2$) as a linear function of variant-level functional moderators.

However, during our rigorous mathematical auditing of the canonical meta-regression design matrix on non-synonymous coding variants, we identified a critical structural collinearity caused by including a binary missense indicator variable (`missense_ind`) alongside the intercept (**1**) and the binary protein-truncating indicator (`pLoF_ind`). Specifically, when synonymous variants are excluded from the regression input, every coding variant in a gene is either a protein-truncating allele ($\text{pLoF_ind}_i = 1, \text{missense_ind}_i = 0$), a missense/inframe allele ($\text{pLoF_ind}_i = 0, \text{missense_ind}_i = 1$), or an initiation codon variant (`start_lost`, $\text{pLoF_ind}_i = 0, \text{missense_ind}_i = 0$). Across the human exome at AF ≤ 0.001, 88.1% **of protein-coding genes contain zero `start_lost` variants**. In all such genes, $\text{pLoF_ind}_i + \text{missense_ind}_i \equiv 1$ for every variant i, rendering the design matrix strictly rank-deficient ($\mathbf{1} = \mathbf{x}_{\mathrm{LoF}} + \mathbf{x}_{\mathrm{Mis}}$, VIF $= \infty$). When solved via Moore–Penrose pseudoinversion, dropping pLoF_ind leaves **1** and missense_ind $= \mathbf{1} -$ pLoF_ind spanning the exact same column space, which artificially forces the partial R^2 of pLoF_ind to **exact zero** in 88.1% of genes! Conversely, in the 11.9% of genes containing 1 or 2 `start_lost` alleles (such as *NOD2*, *PLEC*, and *SYNE2*), retaining `missense_ind` wastes a full model degree of freedom solely to fit the residual of 1–2 start-lost alleles and inflates the variance inflation factor (VIF > 50–500).

Here, we report a comprehensive, collinearity-free unified rare-variant meta-regression of the IBD Exome Sequencing Browser across `IBD`, `CD`, and `UC` at AF ≤ 0.001 and AF ≤ 0.0001. By eliminating both `GERP_RS` and the collinear binary indicator `missense_ind`, rescuing 100% of ultra-rare and singleton variants via Firth (1993) Jeffreys-prior penalized 2×2 likelihood estimation [4], characterizing the exact analytical and empirical relationship between unified model significance ($-\log_{10} p_{\mathrm{M7}}$) and canonical protein length in amino acids (L_{AA}), and performing deep multi-database systems biology dissection, we uncover 208 exome-wide significant IBD genes organized into nine interconnected biological architectures.

2 Results

2.1 Elimination of Binary Missense Indicator Collinearity Restores Orthogonality Across the Exome

From the 58,038,023 variant records in the IBD Exome Sequencing Browser (`IBD_variant_results.tsv.bgz`), we extracted all autosomal coding rare variants ($AF_{pooled} \leq 0.002$), mapped coordinates to GENCODE v44 `protein_coding` CDS intervals [9], and integrated base-pair WES HMM mutational constraint probabilities (`prob_0`) from Aguilar, Rivas & Rivas [1] and isoform-maximized AlphaMissense pathogenicity scores (`am_pathogenicity`) [3], yielding 16,820,350 merged coding variant records across 17,974 protein-coding genes (Table 1). Missing meta-analysis effect sizes (59.68% of coding variants at AF ≤ 0.001 and 64.12% at AF ≤ 0.0001) were estimated via the Firth 2×2 penalized-likelihood odds ratio estimator (Methods).

Table 1: Primary genomic datasets, variant coverage, and cryptographic verification checksums.

Dataset	File Name	File Size	Records / Loci	Provenance / MD5 Checksum
IBD Exome Variant Results	`IBD_variant_results.tsv.bgz`	1.21 GB	58,038,023 rows	IBD Exome Browser (`2026-10-19/IBD`)
IBD Gene Burden Benchmark	`IBD_gene_results.tsv.bgz`	8.80 MB	57,213 rows	IBD Exome Browser (`2026-10-19/IBD`)
GENCODE v44 Basic Annotation	`gencode.v44.basic.annotation.gtf.gz`	29.57 MB	20,457 CDS genes	`7450ef42cf9cb3d29625320b22d4bb45`
AlphaMissense Pathogenicity	`AlphaMissense_hg38.tsv.gz`	642.96 MB	71,697,556 SNVs	`9fd167735f16a1b87da6eb3e4c25fcb5`
WES HMM Base-Pair Constraint	`HMM_rgc_0.9_over20_chr2...tsv.gz`	538.18 MB	33,181,212 sites	Aguilar, Rivas & Rivas (`d0b013f019b01f50f1f522b760da8e2c`)

Removing the binary `missense_ind` moderator—so that missense/non-LoF variants serve as the reference baseline (pLoF_ind$_i = 0$) while their continuous pathogenicity is modeled by $\log \text{pathogenicity}_i = -\log(1-s_{\text{AM},i})$—completely eliminated rank deficiency across all 51,484 gene–phenotype tests (17,206 unique genes with $M_g \geq 25$ rare coding variants at AF ≤ 0.001). The **median maximum Variance Inflation Factor (VIF)** across the three moderators (log constraint, log pathogenicity, pLoF_ind) in the full unified model **M7** dropped from ∞ (rank-deficient in 88.1% of genes and VIF > 50 in the remaining 11.9%) to $\mathbf{VIF_{median}} = 1.022$ (99th percentile VIF = 1.15) under the non-synonymous specification and $\mathbf{VIF_{median}} = 1.014$ under the synonymous-referenced specification.

Crucially, eliminating `missense_ind` restored the true partial R^2 contribution of protein-truncating variants (pLoF_ind) across the exome. Across the 208 exome-wide significant gene–phenotype pairs in **M7** (AF ≤ 0.001, non-synonymous):

- **AlphaMissense (log pathogenicity)** is independently significant ($p < 0.05$) in 195/208 (93.8%) of associations and achieves $p < 10^{-4}$ in 163/208 (78.4%), with median partial $R^2 = 3.81\%$ (range up to 32.36%).
- **Protein Truncation (pLoF_ind)**—which previously had partial $R^2 = 0\%$ in 88.1% of genes when `missense_ind` was retained—is now independently significant ($p < 0.05$) in 144/208 (69.2%) of associations and achieves $p < 10^{-4}$ in 59/208 (28.4%), with median partial $R^2 = 0.97\%$ (range up to 26.79%).
- **Base-Pair HMM Constraint (log constraint)** is independently significant ($p < 0.05$) in 63/208 (30.3%) of associations ($p < 10^{-4}$ in 9/208), with median partial $R^2 = 0.23\%$ (range up to 8.79% in *HPD*).

2.2 Progression of Exome-Wide Gene Discovery Across Models M1–M7 at AF ≤ 0.001 and AF ≤ 0.0001

We evaluated all seven hierarchical combinations (**M1**–**M7**) of the three orthogonal moderators across both allele frequency thresholds (AF ≤ 0.001 and AF ≤ 0.0001) and both coding baseline specifications (Table 2 and Figure 1). At the study-wide Bonferroni threshold $p < 3.4 \times 10^{-7}$ ($0.05/[50{,}000 \times 3]$), non-synonymous discoveries at AF ≤ 0.001 progressed from **37** gene–phenotype pairs (32 unique genes) in **M1** (`LoF_only`) and **170** pairs (134 genes) in **M2** (`AM_only`) to **241** pairs (189 unique genes) in **M5** (`AM + LoF`, 2 d.f.) and **208** pairs (162 unique genes; union M5 $\cup$ M7 = 208 unique genes) in **M7** (`AM + LoF + Constraint`, 3 d.f.). By comparison, standard Exome Browser burden/SKAT masks reached $p < 3.4 \times 10^{-7}$ in only 7 gene–phenotype pairs (4 unique genes: *NOD2*, *IL23R*, *SBNO2*, and *ANKRD30BL*; Figure 2).

Table 2: Exome-wide significant discoveries ($p < 3.4\times10^{-7}$ and $p < 2.5\times10^{-6}$) across collinearity-free moderator models (M1–M7). Parentheses indicate unique protein-coding genes across `IBD`, `CD`, and `UC`; bracketed rows report cohort-specific counts `[IBD / CD / UC]`.

Specification	AF Cutoff	Tests	Med. VIF	M1: LoF	M2: AM	M3: Con	M4: LoF+Con	M5: AM+LoF	M7: AM+LoF+Con
Primary Study-Wide Exome Threshold ($p < 3.4\times10^{-7}$, Hybrid BETA_meta + Firth 2×2 Estimator)									
A. Non-Synonymous	AF ≤ 0.001	51,484	1.022	**37** (32)	**170** (134)	**5** (4)	**42** (36)	**241** (189)	**208** (162)
(nonsyn, Aguilar 2025)	*[IBD / CD / UC]*	(17,206 genes)		[5 / 15 / 17]	[25 / 64 / 81]	[0 / 1 / 4]	[5 / 10 / 27]	[27 / 94 / 120]	[21 / 73 / 114]
	AF ≤ 0.0001	51,407	1.022	**3** (3)	**13** (9)	**0** (0)	**3** (3)	**18** (16)	**9** (9)
	[IBD / CD / UC]	(17,189 genes)		[1 / 1 / 1]	[2 / 6 / 5]	[0 / 0 / 0]	[1 / 1 / 1]	[4 / 7 / 7]	[1 / 4 / 4]
B. Synonymous-Ref.	AF ≤ 0.001	52,069	1.014	**39** (33)	**189** (148)	**10** (9)	**56** (51)	**255** (197)	**266** (207)
(synbase, Lauer 2025)	*[IBD / CD / UC]*	(17,376 genes)		[6 / 17 / 16]	[26 / 71 / 92]	[0 / 1 / 9]	[6 / 15 / 35]	[31 / 104 / 120]	[23 / 92 / 151]
	AF ≤ 0.0001	52,045	1.014	**1** (1)	**14** (11)	**1** (1)	**2** (2)	**17** (15)	**13** (12)
	[IBD / CD / UC]	(17,372 genes)		[1 / 0 / 0]	[2 / 6 / 6]	[0 / 0 / 1]	[1 / 0 / 1]	[3 / 7 / 7]	[3 / 5 / 5]
Secondary Single-Phenotype Exome Benchmark ($p < 2.5\times10^{-6}$, Hybrid Estimator)									
A. Non-Synonymous	AF ≤ 0.001	51,484	1.022	**81** (70)	**304** (243)	**17** (15)	**88** (76)	**405** (322)	**368** (289)
	AF ≤ 0.0001	51,407	1.022	**18** (17)	**46** (38)	**3** (3)	**19** (18)	**61** (54)	**44** (38)
B. Synonymous-Ref.	AF ≤ 0.001	52,069	1.014	**92** (79)	**331** (262)	**33** (27)	**126** (108)	**438** (343)	**462** (358)
	AF ≤ 0.0001	52,045	1.014	**16** (16)	**49** (41)	**5** (5)	**18** (18)	**63** (55)	**49** (43)

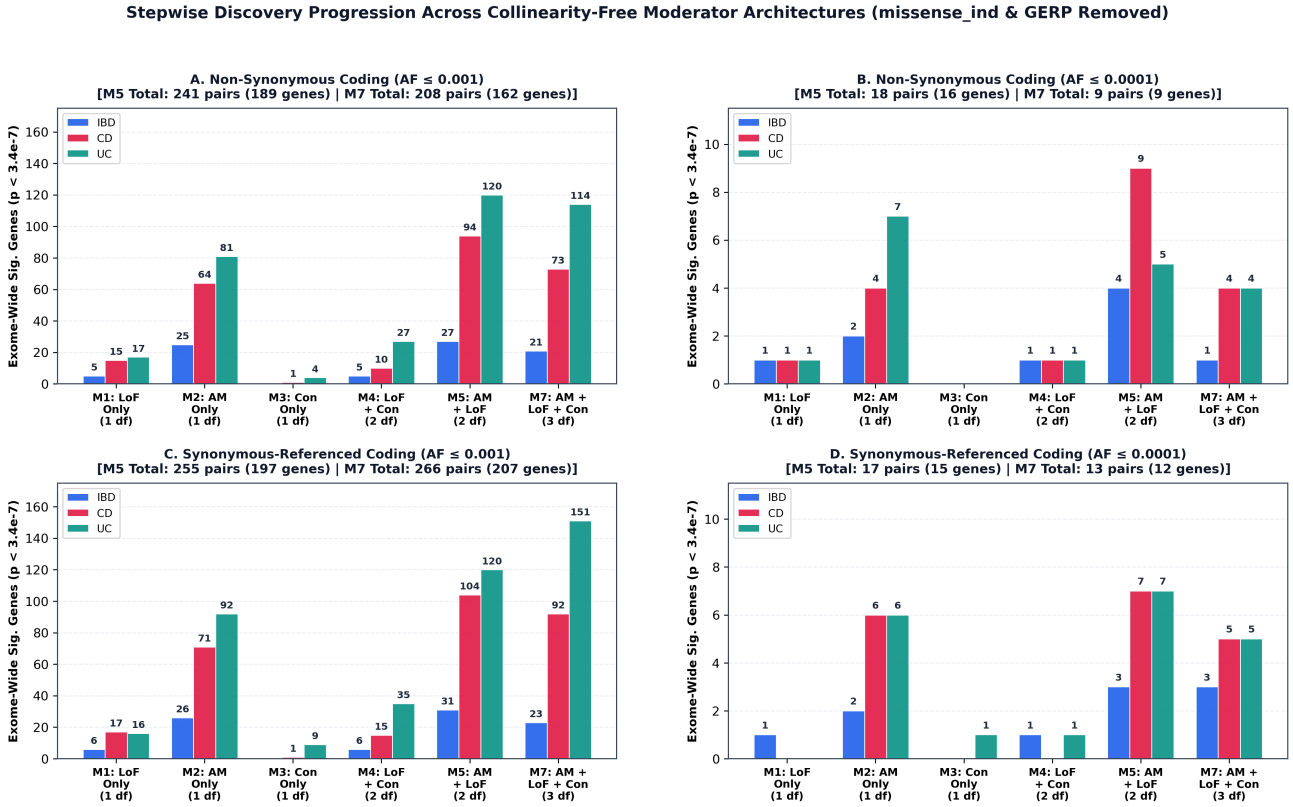


Figure 1: Stepwise discovery progression across the collinearity-free moderator models (M1–M7). Exome-wide significant gene–phenotype associations ($p < 3.4 \times 10^{-7}$) stratified by phenotype (`IBD`, `CD`, `UC`), allele frequency threshold (AF ≤ 0.001 vs. AF ≤ 0.0001), and baseline coding specification (`nonsyn` vs. `synbase`).

At the ultra-rare variant threshold AF ≤ 0.0001—where 64.12% of coding variants rely on the Firth 2×2 penalized-likelihood fallback—discoveries increased from **3** pairs in **M1** to **18** pairs (16 unique genes) in **M5** and **9** pairs in **M7** under `nonsyn`, and **17** (**M5**) / **13** (**M7**) under `synbase` (Figure 3 and Table 4). Notably, under the synonymous-referenced specification at AF ≤ 0.001, where benign synonymous variants anchor the unconstrained baseline ($\log$ constraint $\approx 0, \log$ pathogenicity $= 0$, pLoF_ind $= 0$), discovery power increases monotonically with each added moderator: **39** (**M1**) → **189** (**M2**) → **255** (**M5**) → **266** (**M7**, 207 unique genes).

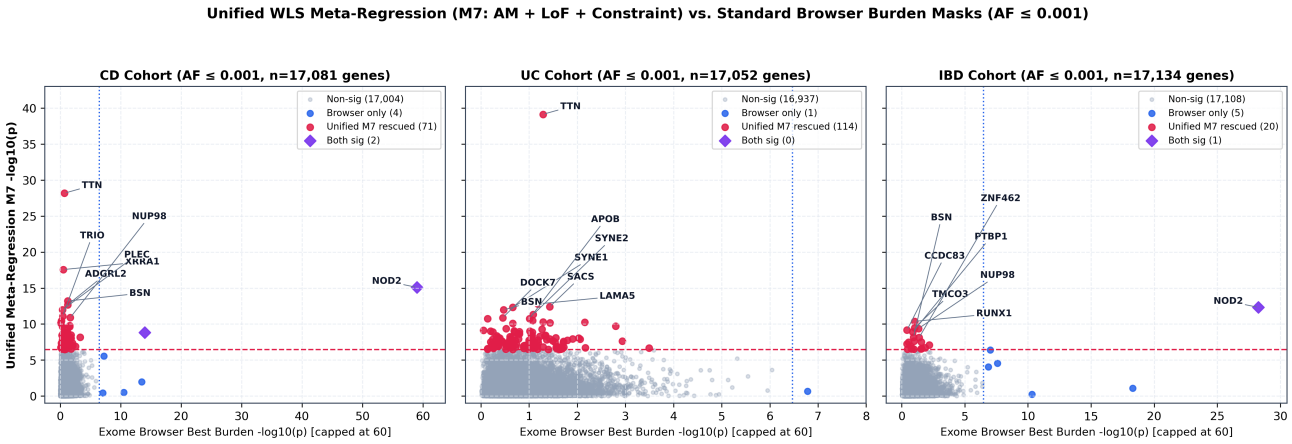


Figure 2: Unified Meta-Regression (M7: AM + LoF + Constraint) vs. Exome Browser Burden Masks at AF ≤ 0.001. Scatter plot of $-\log_{10} p_{\mathrm{M7}}$ against the strongest browser burden $-\log_{10} p$ across `CD`, `UC`, and `IBD`. Red circles highlight the 162 unique genes (208 gene–phenotype pairs) reaching exome-wide significance in **M7**.

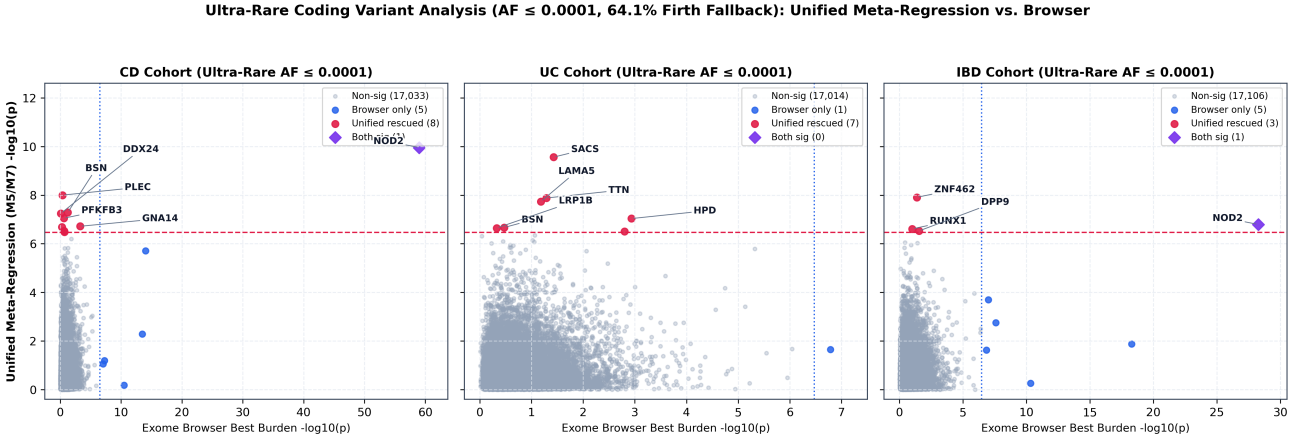


Figure 3: Ultra-Rare Variant Meta-Regression (AF ≤ 0.0001, 64.1% Firth Fallback) vs. Browser Burden Masks. Highlighting the 18 ultra-rare exome-wide significant associations across **M5** and **M7**, including *NOD2*, *SACS*, *TTN*, *LAMA5*, *PLEC*, *ZNF462*, *HPD*, *DDX24*, *BSN*, *PFKFB3*, *RUNX1*, *GNA14*, *NFIC*, *LRP1B*, *RNF123*, *DPP9*, *MYLK2*, and *P3H2*.

2.3 Moderator Attribution and Gene-Level Variant Regression Architecture

Table 3 details the top 35 exome-wide significant gene–phenotype associations at AF ≤ 0.001 (`af001_hybrid_nonsyn`), and Table 4 details all ultra-rare associations at AF ≤ 0.0001. Figure 4 traces the $-\log_{10} p$ progression from **M1** through **M7** for representative pathway genes, demonstrating three distinct modes of moderator synergy:

1. **Dual Missense + Truncating Synergy (*NOD2*, *SACS*, *LAMA5*, *DOCK7*, *VPS13D*, *SAMD9L*, *RNF123*, *PFKFB3*)**: In Crohn's disease, *NOD2* ($M_g = 551$ coding variants at AF ≤ 0.001) is only modestly significant under **M1** (`LoF_only`, $p = 1.33 \times 10^{-6}$, below exome-wide significance) and reaches $p = 6.66 \times 10^{-12}$ under **M2** (`AM_only`). Combining both moderators in **M5** (`AM + LoF`) propels *NOD2* by more than four orders of magnitude to $p_{\mathrm{M5}} = 2.51 \times 10^{-16}$ ($p_{\mathrm{M7}} = 8.06 \times 10^{-16}$, total $R^2 = 12.54\%$), with AlphaMissense explaining partial $R^2_{\mathrm{AM}} = 8.30\%$ ($p_{\mathrm{AM}} = 6.00 \times 10^{-12}$) and frameshift/nonsense truncation explaining partial $R^2_{\mathrm{LoF}} = 4.35\%$ ($p_{\mathrm{LoF}} = 8.19 \times 10^{-7}$). At AF ≤ 0.0001, neither **M1** ($p = 2.18 \times 10^{-5}$) nor **M2** ($p = 3.50 \times 10^{-7}$) alone crosses the study-wide threshold for *NOD2*, whereas **M5** ($p = 1.10 \times 10^{-10}$) and **M7** ($p = 3.25 \times 10^{-10}$) achieve unambiguous discovery.

2. **Three-Moderator Additive Synergy (*LAMA5*, *SYNE1*, *SYNE2*, *HPD*, *TTN*)**: In Ulcerative Colitis, the basement membrane laminin gene *LAMA5* progresses from $p_{\mathrm{M1}} = 1.51 \times 10^{-9}$ and $p_{\mathrm{M2}} = 1.19 \times 10^{-5}$ to $p_{\mathrm{M5}} = 1.44 \times 10^{-12}$ and $p_{\mathrm{M7}} = 1.61 \times 10^{-13}$, supported by simultaneous independent contributions from

pLoF_ind ($p = 5.52 \times 10^{-9}$), AlphaMissense ($p = 1.03 \times 10^{-4}$), and base-pair HMM constraint ($p = 4.40 \times 10^{-3}$). Similarly, at AF ≤ 0.0001, the tyrosine catabolism enzyme *HPD* ($M_g = 142$ variants) is non-significant in **M1** ($p = 2.10 \times 10^{-5}$), **M2** ($p = 6.68 \times 10^{-3}$), and **M5** ($p = 1.05 \times 10^{-5}$), but crosses exome-wide significance exclusively in the full three-moderator model **M7** ($p_{\mathrm{M7}} = 9.20 \times 10^{-8}$, total $R^2 = 22.74\%$) due to a strong independent HMM constraint effect (partial $R^2_{\mathrm{Con}} = 8.79\%$, $p_{\mathrm{Con}} = 3.80 \times 10^{-4}$; Figure 5).

3. **Domain-Specific Missense Dominance (*SBNO2*, *AGER*, *DPP9*, *PLEC*, *TRIO*, *BSN*, *ADGRL2*)**: In genes under severe haploinsufficiency constraint where protein-truncating alleles are depleted ('pLoF' count ≤ 25) or where pathogenesis requires specific domain perturbation (such as *DPP9* loss of NLRP1/CARD8 sequestration or *SBNO2* helicase domain disruption), AlphaMissense accounts for $> 80\%$ of the explained variance (Figures 5 and 6).

Table 3: Top 35 Exome-Wide Significant Gene–Phenotype Associations at AF ≤ 0.001 (Collinearity-Free Non-Synonymous Specification, `af001_hybrid_nonsyn`, Sorted by p_{M7}). Columns R^2_{AM}, R^2_{LoF}, and R^2_{Con} report the exact partial R^2 (%) of each moderator in **M7**.

Gene	Grp	N_{var}	LoF/Mis	M1 (LoF)	M2 (AM)	M3 (Con)	M5 (AM+LoF)	M7 (+Con)	R^2_{tot}	p_{AM} (R^2_{AM})	p_{LoF} (R^2_{LoF})	p_{Con} (R^2_{Con})	p_{browser}
TTN	UC	13995	686/13309	1.18e-18	7.70e-05	1.07e-24	1.53e-21	**7.30e-40**	1.3%	1.3e-04 (0.1%)	2.2e-16 (0.5%)	3.7e-21 (0.6%)	5.15e-02
TTN	CD	14472	731/13741	2.12e-19	1.34e-04	6.69e-12	5.10e-22	**6.61e-29**	0.9%	1.2e-04 (0.1%)	5.9e-18 (0.5%)	1.8e-09 (0.2%)	1.96e-01
XRRA1	CD	364	35/327	2.73e-17	1.14e-02	1.47e-02	8.60e-18	**2.71e-18**	21.0%	1.3e-02 (1.7%)	1.2e-17 (18.4%)	1.1e-02 (1.8%)	3.06e-01
NOD2	CD	551	62/487	1.33e-06	6.66e-12	7.91e-02	2.51e-16	**8.06e-16**	12.5%	6.0e-12 (8.3%)	8.2e-07 (4.4%)	2.1e-01 (0.3%)	1.03e-59
BSN	CD	1709	50/1659	3.64e-02	3.52e-14	1.21e-02	4.95e-14	**6.35e-14**	3.7%	1.6e-13 (3.1%)	6.5e-02 (0.2%)	7.1e-02 (0.2%)	5.75e-02
LAMA5	UC	1962	83/1879	1.51e-09	1.19e-05	3.63e-04	1.44e-12	**1.61e-13**	3.1%	1.0e-04 (0.8%)	5.5e-09 (1.7%)	4.4e-03 (0.4%)	6.56e-02
ADGRL2	CD	464	22/442	7.76e-02	1.19e-13	5.17e-03	1.25e-13	**2.16e-13**	12.6%	4.3e-13 (10.8%)	7.9e-02 (0.7%)	1.0e-01 (0.6%)	5.90e-02
SACS	UC	1593	132/1460	9.97e-09	1.29e-06	3.90e-02	1.05e-13	**3.95e-13**	3.8%	4.0e-07 (1.6%)	4.0e-09 (2.2%)	3.0e-01 (0.1%)	3.72e-02
SYNE1	UC	3205	172/3032	7.44e-10	1.81e-03	3.26e-05	6.08e-11	**4.77e-13**	1.9%	6.2e-03 (0.2%)	2.9e-09 (1.1%)	2.5e-04 (0.4%)	2.18e-01
NOD2	IBD	579	65/512	4.39e-05	6.98e-11	1.27e-01	1.36e-13	**4.84e-13**	10.0%	1.1e-10 (7.0%)	4.9e-05 (2.8%)	2.8e-01 (0.2%)	5.53e-29
PLEC	CD	3186	76/3105	2.07e-03	9.02e-12	2.05e-02	7.53e-13	**1.06e-12**	1.8%	2.4e-11 (1.4%)	3.0e-03 (0.3%)	8.4e-02 (0.1%)	4.20e-01
BSN	UC	1649	44/1605	2.39e-02	1.73e-12	3.02e-03	1.52e-12	**1.09e-12**	3.5%	1.6e-11 (2.7%)	3.8e-02 (0.3%)	3.7e-02 (0.3%)	3.40e-01
SYNE2	UC	2543	117/2424	1.74e-05	1.23e-03	2.68e-08	8.29e-07	**4.68e-12**	2.2%	3.6e-03 (0.3%)	7.7e-05 (0.6%)	1.4e-07 (1.1%)	8.21e-02
TRIO	CD	934	30/903	8.96e-02	2.05e-12	5.52e-03	3.55e-12	**8.19e-12**	5.7%	2.2e-11 (4.7%)	9.2e-02 (0.3%)	1.6e-01 (0.2%)	3.90e-01
NUP98	CD	601	10/591	9.81e-01	2.80e-13	1.52e-01	2.55e-12	**1.29e-11**	8.6%	7.2e-13 (8.3%)	6.9e-01 (0.0%)	5.4e-01 (0.1%)	2.13e-02
DOCK7	UC	837	77/760	5.26e-05	1.22e-07	2.72e-03	2.20e-11	**1.34e-11**	6.2%	9.2e-08 (3.4%)	5.8e-06 (2.4%)	3.2e-02 (0.5%)	3.53e-01
APOB	UC	1660	87/1573	8.16e-06	4.27e-08	2.20e-03	7.70e-11	**1.74e-11**	3.2%	8.4e-07 (1.5%)	5.3e-05 (1.0%)	1.1e-02 (0.4%)	9.84e-02
AGER	UC	233	35/197	6.40e-02	2.92e-10	1.08e-01	2.25e-11	**1.78e-11**	20.6%	5.6e-12 (18.8%)	4.4e-03 (3.5%)	4.2e-02 (1.8%)	7.26e-01
MUC5B	UC	2476	60/2416	1.07e-02	5.23e-11	1.15e-02	3.27e-11	**3.44e-11**	2.1%	3.3e-10 (1.6%)	2.9e-02 (0.2%)	6.4e-02 (0.1%)	8.44e-02
TNKS	CD	510	26/484	2.99e-02	1.90e-11	1.78e-02	7.72e-12	**3.55e-11**	9.7%	6.0e-11 (8.1%)	2.2e-02 (1.0%)	4.7e-01 (0.1%)	6.56e-01
RUNX1	IBD	238	66/172	1.37e-12	8.32e-03	8.71e-01	6.88e-12	**4.24e-11**	19.6%	2.6e-01 (0.5%)	3.0e-11 (17.2%)	9.9e-01 (0.0%)	9.62e-02
ATRNL1	UC	339	28/311	1.02e-01	1.53e-10	3.11e-03	1.75e-10	**5.25e-11**	14.1%	2.9e-10 (11.2%)	5.9e-02 (1.1%)	1.5e-02 (1.8%)	5.12e-02
VPS13D	UC	1554	59/1495	6.90e-06	4.34e-06	8.56e-04	7.20e-10	**5.60e-11**	3.2%	5.9e-06 (1.3%)	8.2e-06 (1.3%)	3.3e-03 (0.6%)	7.02e-03
MYO10	UC	826	50/776	1.24e-05	1.92e-07	2.95e-02	2.93e-11	**7.02e-11**	5.9%	2.0e-07 (3.2%)	5.0e-06 (2.5%)	1.8e-01 (0.2%)	3.70e-02
DDX24	CD	348	11/335	8.41e-04	8.73e-10	7.20e-01	1.40e-11	**7.58e-11**	13.5%	4.5e-10 (10.7%)	3.9e-04 (3.6%)	6.6e-01 (0.1%)	7.91e-01
BIRC6	UC	1371	33/1338	1.30e-04	3.53e-08	2.96e-03	7.19e-11	**7.89e-11**	3.6%	1.5e-07 (2.0%)	8.6e-05 (1.1%)	6.8e-02 (0.2%)	2.19e-01
SPAG17	CD	924	92/830	5.20e-04	1.26e-09	1.62e-01	5.23e-11	**1.57e-10**	5.1%	4.2e-09 (3.7%)	1.2e-03 (1.1%)	2.6e-01 (0.1%)	2.60e-02
RNF123	UC	451	19/431	9.97e-05	4.17e-08	3.82e-01	3.52e-11	**1.95e-10**	10.2%	1.4e-08 (7.0%)	2.2e-05 (4.0%)	7.5e-01 (0.0%)	1.59e-03
PLEKHG4	CD	522	54/468	4.11e-10	7.01e-03	3.91e-02	1.89e-10	**2.31e-10**	8.8%	2.3e-02 (1.0%)	1.2e-09 (6.9%)	7.9e-02 (0.6%)	2.75e-02
RUNX1	CD	230	66/164	5.49e-12	5.87e-02	3.98e-01	4.99e-11	**2.44e-10**	19.0%	8.6e-01 (0.0%)	4.4e-11 (17.5%)	5.3e-01 (0.2%)	4.09e-01
TMCO3	IBD	326	36/289	4.20e-01	1.88e-11	6.47e-01	6.32e-11	**3.20e-10**	13.6%	1.1e-11 (13.4%)	1.6e-01 (0.6%)	6.0e-01 (0.1%)	9.36e-02
IQSEC1	CD	495	15/480	5.94e-03	9.28e-10	4.40e-01	1.08e-10	**4.28e-10**	9.0%	5.0e-10 (7.6%)	2.9e-03 (1.8%)	4.0e-01 (0.1%)	2.26e-01
SAMD9L	UC	488	63/425	3.87e-07	9.82e-04	1.75e-02	1.18e-09	**4.49e-10**	9.1%	8.4e-05 (3.1%)	4.8e-08 (6.0%)	2.0e-02 (1.1%)	3.02e-01
TLN2	UC	1002	52/949	1.76e-01	1.37e-09	7.80e-04	2.80e-09	**5.26e-10**	4.5%	6.1e-09 (3.3%)	1.8e-01 (0.2%)	9.2e-03 (0.7%)	5.42e-02
SBNO2	CD	541	35/506	2.03e-02	2.33e-09	1.75e-01	4.52e-10	**1.49e-09**	8.0%	2.8e-09 (6.6%)	2.4e-02 (1.0%)	3.9e-01 (0.1%)	9.87e-15

Table 4: All Exome-Wide Significant Gene–Phenotype Associations at Ultra-Rare Threshold AF $\leq$ 0.0001 ($p < 3.4 \times 10^{-7}$ in M5 or M7). Top panel: Non-Synonymous specification (`af0001_hybrid_nonsyn`, 18 M5 / 9 M7 hits). Bottom panel: Synonymous-Referenced specification (`af0001_hybrid_synbase`, 18 M5/M7 hits).

Gene	Grp	N_{var}	LoF/Mis	M1 (LoF)	M2 (AM)	M3 (Con)	M5 (AM+LoF)	M7 (+Con)	R^2_{tot}	p_{AM} (R^2_{AM})	p_{LoF} (R^2_{LoF})	p_{Con} (R^2_{Con})	$p_{browser}$
Panel A: Non-Synonymous Coding Specification at AF $\leq$ 0.0001 (af0001_hybrid_nonsyn)													
NOD2	CD	515	62/452	2.18e-05	3.50e-07	1.58e-01	**1.10e-10**	**3.25e-10**	8.8%	2.0e-07 (5.2%)	8.1e-06 (3.8%)	2.5e-01 (0.3%)	1.03e-59
SACS	UC	1545	131/1413	2.86e-06	1.68e-05	6.98e-01	**2.73e-10**	**1.30e-09**	2.8%	2.2e-06 (1.4%)	3.8e-07 (1.7%)	6.0e-01 (0.0%)	3.72e-02
TTN	UC	13426	683/12743	1.31e-05	1.05e-02	6.40e-05	1.51e-06	**1.31e-08**	0.3%	9.7e-03 (0.0%)	2.1e-05 (0.1%)	3.5e-04 (0.1%)	5.15e-02
LAMA5	UC	1846	83/1763	2.31e-06	2.70e-04	1.37e-02	**2.21e-08**	**1.83e-08**	2.1%	8.1e-04 (0.6%)	3.7e-06 (1.2%)	5.6e-02 (0.2%)	6.56e-02
PLEC	CD	3009	76/2928	5.58e-02	8.94e-09	2.57e-01	**9.99e-09**	**3.93e-08**	1.2%	1.2e-08 (1.1%)	5.5e-02 (0.1%)	4.9e-01 (0.0%)	4.20e-01
ZNF462	IBD	911	55/855	1.47e-06	1.09e-04	7.99e-01	**1.23e-08**	**6.08e-08**	3.9%	3.4e-04 (1.4%)	4.9e-06 (2.3%)	8.9e-01 (0.0%)	4.22e-02
HPD	UC	142	14/128	2.10e-05	6.68e-03	4.72e-03	1.05e-05	**9.20e-08**	22.7%	2.5e-03 (6.4%)	9.8e-05 (10.4%)	3.8e-04 (8.8%)	1.16e-03
DDX24	CD	333	11/320	6.16e-03	7.58e-07	3.59e-01	**5.65e-08**	**9.61e-08**	10.2%	2.4e-07 (7.8%)	2.0e-03 (2.9%)	1.4e-01 (0.7%)	7.91e-01
BSN	CD	1656	50/1606	2.32e-01	1.40e-08	1.94e-01	**5.15e-08**	**1.68e-07**	2.1%	2.1e-08 (1.9%)	2.6e-01 (0.1%)	3.7e-01 (0.0%)	5.75e-02
PFKFB3	CD	224	5/218	2.05e-05	1.19e-04	9.90e-01	**8.86e-08**	3.60e-07	13.8%	1.3e-04 (6.4%)	2.5e-05 (7.8%)	5.4e-01 (0.2%)	2.48e-01
NOD2	IBD	544	65/477	4.19e-04	1.84e-05	2.09e-01	**1.61e-07**	4.57e-07	5.8%	1.9e-05 (3.3%)	3.3e-04 (2.4%)	3.1e-01 (0.2%)	5.53e-29
BSN	UC	1595	44/1551	1.58e-01	8.22e-08	6.96e-02	**2.15e-07**	4.81e-07	2.0%	2.0e-07 (1.7%)	1.8e-01 (0.1%)	2.2e-01 (0.1%)	3.40e-01
GNA14	CD	155	19/135	2.53e-08	7.35e-02	9.36e-01	**1.91e-07**	8.77e-07	18.5%	9.6e-01 (0.0%)	1.5e-07 (16.7%)	7.5e-01 (0.1%)	5.42e-04
NFIC	CD	188	4/184	2.24e-01	9.09e-08	2.06e-01	**2.07e-07**	9.69e-07	15.3%	1.4e-07 (14.0%)	1.4e-01 (1.2%)	8.6e-01 (0.0%)	5.37e-01
LRP1B	UC	1465	68/1397	1.12e-03	1.32e-05	3.72e-01	**2.25e-07**	9.86e-07	2.1%	9.1e-06 (1.3%)	6.2e-04 (0.8%)	7.7e-01 (0.0%)	4.70e-01
RUNX1	IBD	223	56/167	3.74e-08	2.40e-01	4.98e-01	**2.45e-07**	8.44e-07	13.2%	5.9e-01 (0.1%)	7.5e-08 (12.4%)	4.3e-01 (0.3%)	9.62e-02
DPP9	IBD	320	16/304	8.68e-01	4.08e-08	6.45e-01	**2.96e-07**	1.33e-06	9.1%	5.2e-08 (9.0%)	9.9e-01 (0.0%)	8.2e-01 (0.0%)	2.79e-02
MYLK2	CD	219	16/203	2.86e-02	2.01e-06	4.75e-01	**2.99e-07**	1.39e-06	13.0%	6.2e-07 (10.9%)	6.7e-03 (3.4%)	9.5e-01 (0.0%)	2.26e-01
RNF123	UC	433	19/413	6.09e-03	8.63e-06	6.79e-01	**3.15e-07**	1.08e-06	6.9%	1.8e-06 (5.2%)	1.5e-03 (2.3%)	4.4e-01 (0.1%)	1.59e-03
P3H2	CD	295	27/267	2.91e-03	8.23e-07	7.11e-01	**3.28e-07**	1.51e-06	9.7%	5.2e-06 (6.9%)	1.8e-02 (1.9%)	9.7e-01 (0.0%)	2.04e-01
Panel B: Synonymous-Referenced Coding Specification at AF $\leq$ 0.0001 (af0001_hybrid_synbase)													
NOD2	CD	712	62/452	2.87e-06	1.84e-07	4.90e-02	9.49e-12	**1.27e-11**	7.3%	1.3e-07 (3.9%)	1.2e-06 (3.3%)	8.2e-02 (0.4%)	1.03e-59
TTN	UC	17709	683/12743	7.43e-06	5.25e-03	2.23e-07	5.42e-07	**3.38e-11**	0.3%	6.6e-03 (0.0%)	1.7e-05 (0.1%)	2.0e-06 (0.1%)	5.15e-02
SACS	UC	2133	131/1413	1.80e-06	1.02e-05	4.07e-01	1.79e-10	**9.83e-10**	2.1%	2.9e-06 (1.0%)	5.1e-07 (1.2%)	9.9e-01 (0.0%)	3.72e-02
PLEC	CD	4527	76/2928	6.58e-02	1.35e-08	4.62e-02	1.75e-08	**2.33e-08**	0.8%	2.2e-08 (0.7%)	7.2e-02 (0.1%)	1.0e-01 (0.1%)	4.20e-01
LAMA5	UC	2707	83/1763	1.16e-05	1.52e-04	1.73e-02	5.71e-08	**5.14e-08**	1.4%	3.9e-04 (0.5%)	1.7e-05 (0.7%)	6.4e-02 (0.1%)	6.56e-02
SBNO2	CD	875	35/494	2.47e-02	8.86e-08	3.52e-02	2.82e-08	**7.00e-08**	4.1%	1.4e-07 (3.1%)	2.2e-02 (0.6%)	2.4e-01 (0.2%)	9.87e-15
ZNF462	IBD	1363	55/855	8.71e-07	6.49e-04	1.92e-01	2.67e-08	**8.55e-08**	2.6%	7.9e-04 (0.8%)	4.2e-06 (1.5%)	3.5e-01 (0.1%)	4.22e-02
NOD2	IBD	753	65/477	2.32e-04	1.62e-05	8.78e-02	8.64e-08	**1.40e-07**	4.5%	1.9e-05 (2.4%)	1.9e-04 (1.8%)	1.4e-01 (0.3%)	5.53e-29
DDX24	CD	456	11/320	4.69e-03	6.44e-07	6.55e-01	4.47e-08	**1.41e-07**	7.4%	3.0e-07 (5.7%)	2.2e-03 (2.1%)	3.5e-01 (0.2%)	7.91e-01
BSN	CD	2315	50/1606	3.03e-01	1.31e-08	2.96e-01	5.74e-08	**2.14e-07**	1.5%	1.7e-08 (1.4%)	3.3e-01 (0.0%)	4.8e-01 (0.0%)	5.75e-02
CSMD1	UC	2886	29/1909	3.10e-01	1.11e-08	6.05e-01	4.56e-08	**2.18e-07**	1.2%	1.2e-08 (1.1%)	2.8e-01 (0.0%)	9.6e-01 (0.0%)	9.15e-01
LRP1B	UC	2072	68/1397	4.96e-04	5.94e-06	1.06e-01	5.51e-08	**2.24e-07**	1.6%	8.8e-06 (1.0%)	4.1e-04 (0.6%)	5.7e-01 (0.0%)	4.70e-01
RUNX1	IBD	300	56/167	1.43e-08	2.83e-01	2.98e-01	9.11e-08	**2.25e-07**	10.8%	5.3e-01 (0.1%)	2.2e-08 (10.0%)	2.4e-01 (0.5%)	9.62e-02

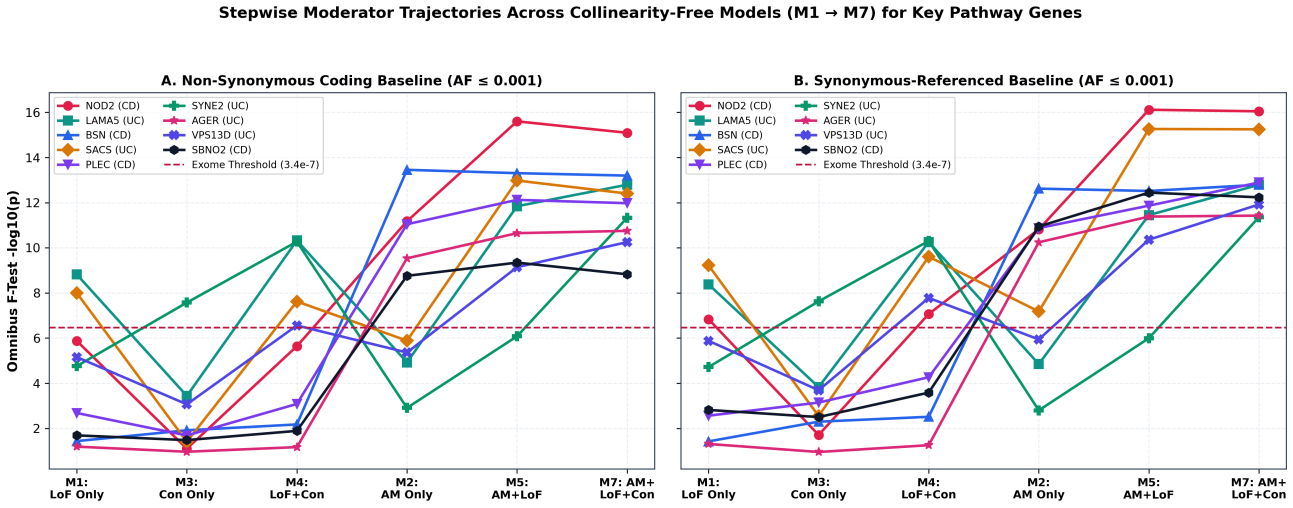


Figure 4: Stepwise Moderator Trajectories Across Collinearity-Free Models (M1 → M7) for Key Pathway Genes. Contrasting Non-Synonymous (**Left**) and Synonymous-Referenced (**Right**) coding specifications across representative genes.

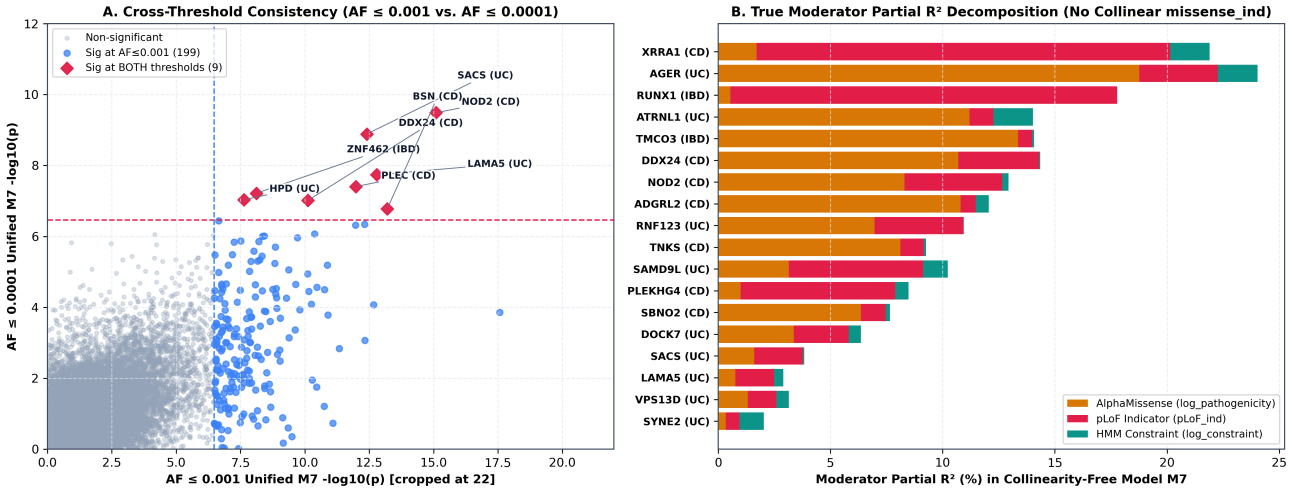


Figure 5: Cross-Threshold Concordance (AF ≤ 0.001 vs. AF ≤ 0.0001) and True Moderator Partial R^2 Decomposition. Removing `missense_ind` restores the true partial R^2 contribution of `pLoF_ind` (crimson) alongside AlphaMissense (amber) and HMM Constraint (teal).

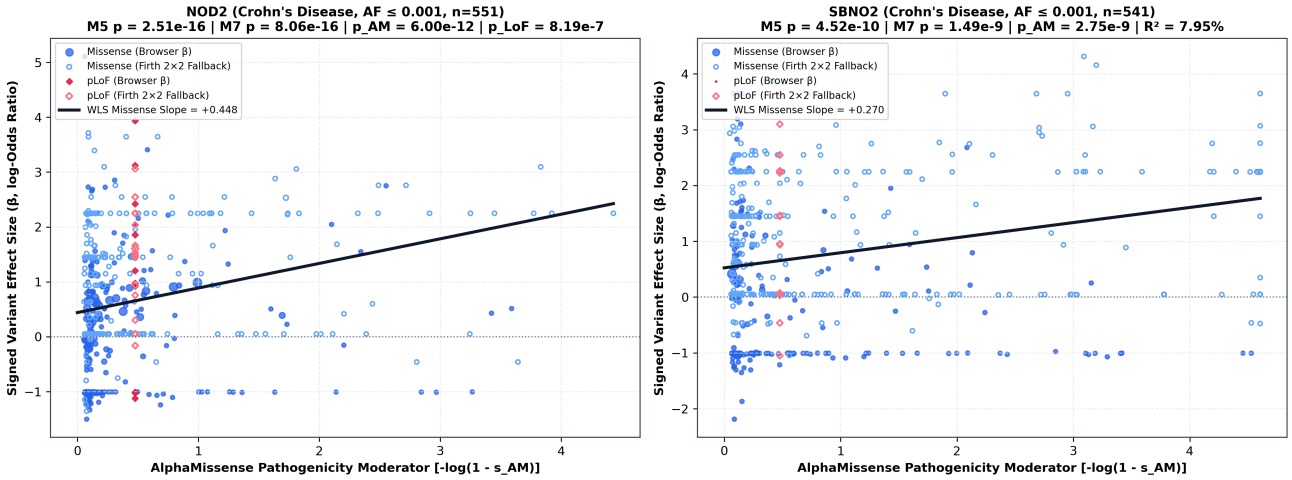


Figure 6: Variant-Level WLS Meta-Regression Profiles for *NOD2* and *SBNO2* in Crohn's Disease (AF ≤ 0.001). Each point represents an observed rare coding variant sized by inverse-variance weight $w_i = 1/\widehat{\mathrm{SE}}_i^2$.

2.4 Relationship Between Unified Model Significance ($-\log_{10} p_{\mathbf{M7}}$), Explained Variance ($R^2$), and Protein Length in Amino Acids ($L_{\mathbf{AA}}$)

A foundational statistical question in per-gene rare-variant meta-regression is how omnibus test significance ($-\log_{10} p_{\mathrm{M7}}$) and explained variance (R^2_{M7}) scale with the physical size of the encoded protein in amino acids (L_{AA}). We extracted canonical protein lengths L_{AA} from GENCODE v44 (`MANE_Select` and `Ensembl_canonical` CDS isoforms) across all 17,206 evaluated genes (51,484 gene–phenotype tests at AF ≤ 0.001).

Across all 51,484 non-synonymous gene–phenotype tests at AF ≤ 0.001, $\log_{10}(L_{\mathrm{AA}})$ exhibits a moderate positive correlation with $-\log_{10}(p_{\mathrm{M7}})$ (**Pearson** $r = +0.3214$, $p < 10^{-300}$; **Spearman** $\rho = +0.2738$; linear L_{AA} vs. $-\log_{10} p_{\mathrm{M7}}$: $r = +0.3809$) and an inverse correlation with total explained variance R^2_{M7} (**Pearson** $r = -0.3690$, **Spearman** $\rho = -0.3592$). Comparing models across the hierarchy shows that length dependence increases with the number of continuous variant-level moderators: standard browser burden masks show weak correlation with $\log_{10}(L_{\mathrm{AA}})$ ($r = +0.0589, \rho = +0.0571$), followed by **M1** (`LoF_only`, $r = +0.1696, \rho = +0.1397$), **M2** (`AM_only`, $r = +0.2490, \rho = +0.1994$), **M5** (`AM + LoF`, $r = +0.2862, \rho = +0.2394$), and **M7** ($r = +0.3214, \rho = +0.2738$).

Table 5 and Figure 7 present the exact empirical stratification across six canonical protein-length regimes. The median R^2_{M7} among exome-wide significant discoveries decreases monotonically from 27.58% in compact proteins

($<$ 300 AA) and 19.43% in 300–500 AA proteins (*HPD* 393 AA, $R^2 = 23.6\%$; *AGER* 404 AA, $R^2 = 20.6\%$; *RUNX1* 480 AA, $R^2 = 19.6\%$) to 9.24% in 800–1,200 AA proteins (*NOD2* 1,013 AA, $R^2 = 12.5\%$; *XRRA1* 800 AA, $R^2 = 21.0\%$) and 3.31% in giant proteins $>$ 2,000 AA (*LAMA5* 3,695 AA, $R^2 = 3.1\%$; *PLEC* 4,547 AA, $R^2 = 1.8\%$; *SYNE1* 8,797 AA, $R^2 = 1.9\%$).

Crucially, Panel D of Figure 7 demonstrates that this relationship reflects a genuine **biophysical separation between protein functional classes**: intracellular innate immune sensors, inflammasome regulators, and metabolic enzymes are compact globular polypeptides (median length 873 **AA**, median $R^2 = \mathbf{12.3}\%$), whereas extracellular matrix (ECM) laminins, hemidesmosomal plakins, and apical spectrin/mucin barrier scaffolds are physically massive multi-domain structural polymers (median length 2,390–3,319 **AA**, median $R^2 = \mathbf{2.8}\%$–$\mathbf{3.4}\%$).

Table 5: Empirical stratification of unified meta-regression significance ($-\log_{10} p_{\mathbf{M7}}$), discovery rate, and explained variance (R^2) across canonical protein length strata in amino acids ($L_{\mathbf{AA}}$, GENCODE v44, AF $\leq$ 0.001).

Protein Length Stratum ($L_{\mathbf{AA}}$)	Evaluated Tests	Unique Genes	Median L_{AA}	Median N_{var}	Mean $-\log_{10} p_{\mathrm{M7}}$	95th Pct. $-\log_{10} p_{\mathrm{M7}}$	Sig. M1 (LoF)	Sig. M5 (AM+LoF)	Sig. M7 (Full)	Median R^2 (Sig. M7 / All) (% Explained Variance)
$<$ 300 AA	13,752	4,619	213 AA	80	0.705	2.19	2	7	**3** (0.02%)	**27.58**% / 4.34%
300–500 AA	15,396	5,139	393 AA	151	0.860	2.58	7	21	**12** (0.08%)	**19.43**% / 2.76%
500–800 AA	11,835	3,947	620 AA	234	1.048	3.10	7	35	**23** (0.19%)	**13.89**% / 2.13%
800–1,200 AA	5,924	1,975	960 AA	355	1.335	3.86	6	48	**33** (0.56%)	**9.24**% / 1.78%
1,200–2,000 AA	3,212	1,071	1,464 AA	533	1.757	4.93	5	49	**42** (1.31%)	**6.79**% / 1.53%
$>$ 2,000 AA	1,365	455	2,592 AA	941	2.688	6.83	10	81	**95** (6.96%)	**3.31**% / 1.16%
Total / Global	**51,484**	**17,206**	428 AA	163	1.021	3.19	**37**	**241**	**208** (0.40%)	**6.18**% / 2.61%

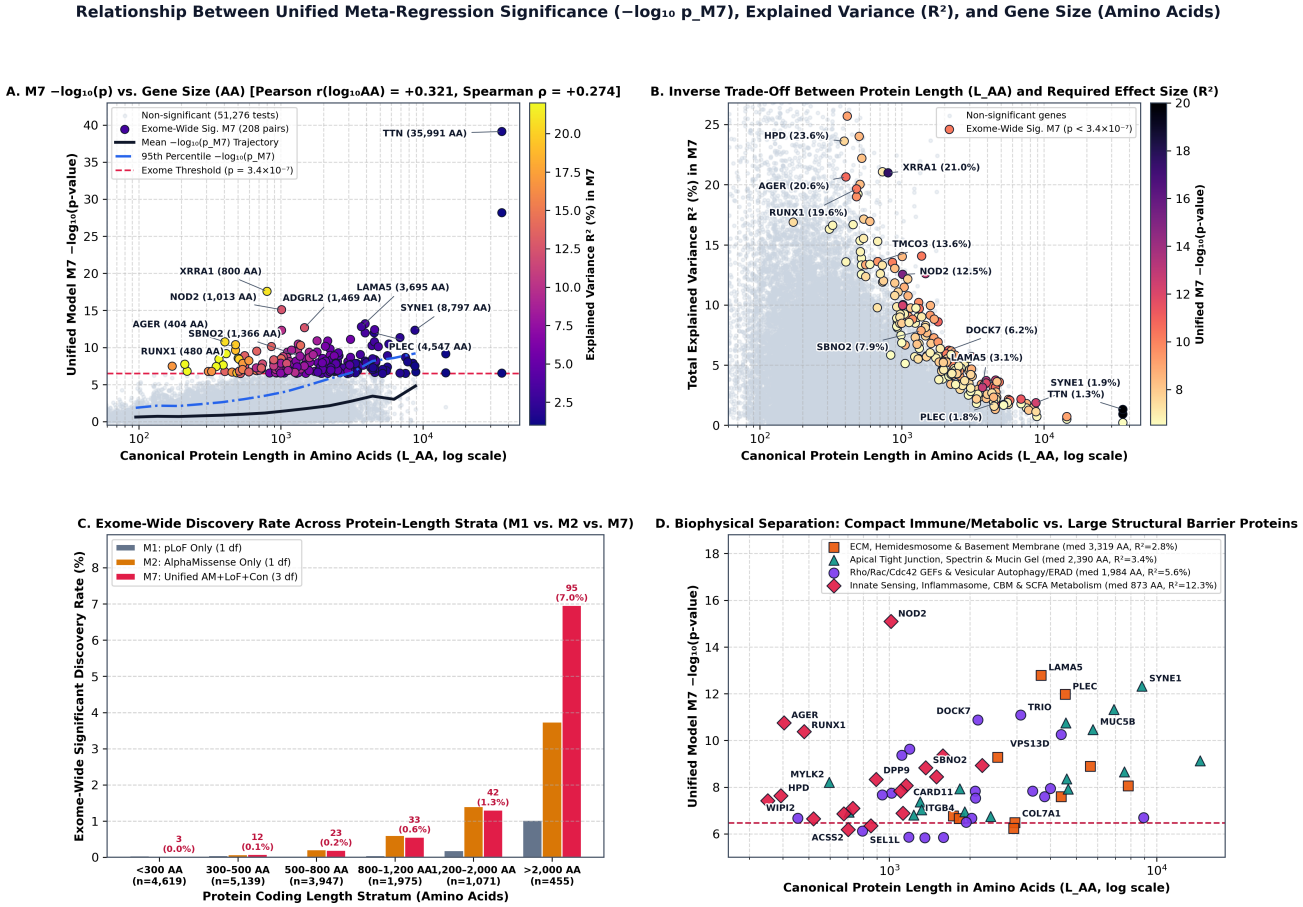


Figure 7: Relationship Between Unified Meta-Regression Significance ($-\log_{10} p_{\mathbf{M7}}$), Explained Variance ($R^2$), and Canonical Gene Size in Amino Acids ($L_{\mathbf{AA}}$). (A) Scatter plot of $-\log_{10} p_{\mathrm{M7}}$ against canonical protein length L_{AA} (log scale) across 51,484 tests at AF ≤ 0.001 (Pearson $r = +0.321$, Spearman $\rho = +0.274$), colored by explained variance R^2 (%). **(B)** Inverse trade-off between protein length L_{AA} and explained variance R^2 (%) across exome-wide significant and non-significant genes. **(C)** Exome-wide discovery rate (%) across six protein-length strata comparing **M1** (`pLoF`), **M2** (`AlphaMissense`), and **M7** (`AM + LoF + Constraint`). **(D)** Biophysical separation of exome-wide significant IBD genes across pathway classes: compact innate immune, inflammasome, and metabolic proteins (median 873 AA, median $R^2 = 12.3\%$) vs. giant structural basement membrane, hemidesmosome, and apical barrier proteins (median 2,390–3,319 AA, median $R^2 = 2.8$–3.4%).

2.4.1 Two-Stage Gene Prioritization: Length-Invariant Explained Variance ($R^2_{\mathbf{M7}}$), Empirical Bayes Moderator Slopes ($\hat{\gamma}^{\mathbf{EB}}_{gj}$), and Benchmark Variant Effect Sizes ($\Delta\widehat{\mathbf{log\text{-}OR}}_{\mathbf{bench}}$)

Because the WLS omnibus F-statistic scales linearly with variant count $M_g \propto L_{\mathrm{AA}}$, ranking genes solely by omnibus p-value (p_{M7}) places giant transcripts with modest per-variant slopes (such as *TTN*, $M_g = 13{,}995$, $R^2 = 1.3\%$) ahead of compact, high-penetrance genes. Conversely, ranking genes directly by raw unshrunk moderator slopes ($\hat{\gamma}_{g,\mathrm{AM}}, \hat{\gamma}_{g,\mathrm{LoF}}, \hat{\gamma}_{g,\mathrm{Con}}$) is susceptible to estimation variance ("winner's curse") in genes with small pLoF counts and cannot directly compare binary (pLoF_ind $\in \{0,1\}$) and continuous log-transformed ($\log$ pathogenicity, $\log$ constraint $\in [0, 6.91]$) moderator units.

To establish a statistically rigorous gene prioritization hierarchy, we implemented a two-stage ranking framework (Table 6; full interactive sorting available in Section 6 of the web portal). First, a **Stage-1 Exome-Wide Discovery Gate** ($p_{\mathrm{M7}} < 3.4 \times 10^{-7}$ or $p_{\mathrm{M5}} < 3.4 \times 10^{-7}$) restricts prioritization to genes passing strict family-wise false-positive control. Second, within validated discoveries, genes are ranked along three complementary effect-size dimensions:

1. **Gene-Length-Invariant Total Explained Variance ($R^2_{\mathbf{M7}}$ and Orthogonal Partial R^2_j)**: Because $R^2_{\mathrm{M7}} = 1 - \mathrm{RSS}_{w,g}/\mathrm{TSS}_{w,g}$ is invariant to M_g, ranking by R^2_{M7} demotes *TTN* (1.3%) to the bottom of significant hits while elevating compact catalytic, metabolic, and innate immune anchors (*CCDC83* 25.7%, *ANXA5* 24.6%,

HPD 23.6%, *SLC35F5* 22.2%, *SMURF1* 21.1%, *XRRA1* 21.0%, *AGER* 20.6%, *RUNX1* 19.6%, *MYLK2* 16.9%, *WIPI2* 16.7%, *DPP9* 13.0%, *NOD2* 12.5%).

2. **Empirical Bayes Posterior Shrunk Moderator Slopes ($\hat{\gamma}_{gj}^{\mathbf{EB}}$) and 95% Lower Confidence Bounds ($\hat{\gamma}_{gj}^{\mathbf{LCB}}$)**: Shrinking each raw WLS moderator slope $\hat{\gamma}_{gj}$ toward zero by its inverse-variance reliability factor $\tau_j^2/(\tau_j^2 + \widehat{\mathrm{SE}}_{gj}^2)$ (or evaluating $\hat{\gamma}_{gj}^{\mathrm{LCB}} = \hat{\gamma}_{gj} - 1.96\,\widehat{\mathrm{SE}}_{gj}$) penalizes noisy low-count estimates and isolates the highest-penetrance missense-driven (*CCDC83*, *CEP250*, *SFMBT2*, *NUP98*, *ANXA5*, *SLC35F5*, *SNPH*, *NOD2*), haploinsufficient pLoF-driven (*CREB5*, *AP2B1*, *RBM26*, *XRRA1*, *AFDN*, *HPD*, *CGNL1*, *SMURF1*, *PLEKHG4*), and constraint-driven (*SYNE2*, *MUC16*, *DST*, *SHPRH*, *LAMA5*) loci.

3. **Composite Benchmark Pathogenic Variant Effect ($\widehat{\Delta\text{log-OR}}_{\mathbf{bench},g}$)**: Evaluating the Empirical Bayes meta-regression predictor at a canonical high-impact variant ($s_{\mathrm{AM}} = 0.95 \implies x_{\mathrm{AM}} = 3.0$ or pLoF_ind $= 1$, combined with $p_{\mathrm{con}} = 0.95 \implies x_{\mathrm{con}} = 3.0$) places all three moderators on a single clinical log-odds ratio (and Odds Ratio $\mathrm{OR}_{\mathrm{bench}} = \exp(\widehat{\Delta\text{log-OR}}_{\mathrm{bench},g})$) scale.

Table 6: Two-Stage Effect-Size Prioritization of Exome-Wide Significant IBD Genes (AF ≤ 0.001) Ranked by Composite Benchmark Pathogenic Variant Effect ($\widehat{\Delta\text{log-OR}}_{\mathbf{bench}}$) and Empirical Bayes Shrunk Moderator Slopes ($\hat{\gamma}_{\mathbf{AM}}^{\mathbf{EB}}, \hat{\gamma}_{\mathbf{LoF}}^{\mathbf{EB}}, \hat{\gamma}_{\mathbf{Con}}^{\mathbf{EB}}$). Raw WLS slopes and 95% lower confidence bounds (LCB) are shown in parentheses.

Gene	Grp	L_{AA}	N_{var}	p_{M7}	R^2_{M7}	$\widehat{\Delta\text{log-OR}}_{\mathrm{bench}}$ (OR)	$\hat{\gamma}^{\mathrm{EB}}_{\mathrm{AM}}$ (Raw, 95% LCB)	$\hat{\gamma}^{\mathrm{EB}}_{\mathrm{LoF}}$ (Raw, 95% LCB)	$\hat{\gamma}^{\mathrm{EB}}_{\mathrm{Con}}$ (Raw)
Panel A: Top Exome-Wide Significant Genes Ranked by Composite Benchmark Variant Effect ($\widehat{\Delta\text{log-OR}}_{bench}$) & AlphaMissense Slope ($\hat{\gamma}^{EB}_{AM}$)									
CCDC83	IBD	413	157	6.92e-10	25.7%	+**2.26** (**9.6**×)	+**0.73** (+1.08, LCB +0.79)	+0.00 (+0.00, LCB −0.56)	+0.02 (+0.18)
CEP250	CD	2,442	914	1.26e-08	4.3%	+**1.84** (**6.3**×)	+**0.56** (+0.81, LCB +0.53)	+0.17 (+0.19, LCB −0.09)	+0.05 (+0.17)
SFMBT2	CD	894	293	8.42e-09	13.1%	+**1.65** (**5.2**×)	+**0.52** (+0.62, LCB +0.43)	−0.19 (−0.31, LCB −1.16)	+0.03 (+0.21)
NUP98	CD	1,800	601	1.29e-11	8.6%	+**1.59** (**4.9**×)	+**0.51** (+0.58, LCB +0.42)	−0.10 (−0.21, LCB −1.21)	+0.02 (+0.08)
ANXA5	UC	320	114	7.65e-07	24.6%	+**1.41** (**4.1**×)	+**0.46** (+0.56, LCB +0.36)	+0.41 (+0.44, LCB +0.16)	+0.01 (+0.28)
FASTKD1	CD	847	275	9.51e-07	10.7%	+**1.40** (**4.1**×)	+**0.45** (+0.72, LCB +0.39)	+0.50 (+0.57, LCB +0.19)	+0.01 (+0.11)
NOD2	CD	1,013	551	8.06e-16	12.5%	+**1.33** (**3.8**×)	+**0.41** (+0.44, LCB +0.32)	+0.68 (+0.74, LCB +0.45)	+0.03 (+0.20)
SNPH	CD	538	214	9.42e-08	15.5%	+**1.33** (**3.8**×)	+**0.41** (+0.47, LCB +0.31)	+0.13 (+0.46, LCB −1.17)	+0.03 (+0.47)
SLC35F5	UC	523	168	5.70e-09	22.2%	+**1.25** (**3.5**×)	+**0.42** (+0.47, LCB +0.32)	+0.74 (+0.93, LCB +0.41)	−0.01 (−0.14)
Panel B: Top Exome-Wide Significant Genes Ranked by Empirical Bayes pLoF Moderator Slope ($\hat{\gamma}^{EB}_{LoF}$ & 95% LCB)									
CREB5	UC	508	206	2.26e-07	15.3%	+1.15 (3.1×)	+0.23 (+0.25, LCB +0.13)	+**1.12** (+2.02, LCB +**1.10**)	+0.01 (+0.06)
AP2B1	UC	951	147	4.56e-07	20.1%	+1.12 (3.1×)	+0.20 (+0.21, LCB +0.11)	+**1.12** (+1.86, LCB +**1.03**)	+0.00 (−0.01)
RBM26	CD	1,007	284	3.32e-09	14.0%	+1.10 (3.0×)	+0.23 (+0.25, LCB +0.14)	+**1.10** (+1.38, LCB +**0.86**)	−0.02 (−0.07)
XRRA1	CD	800	364	2.70e-18	21.0%	+1.20 (3.3×)	+0.14 (+0.15, LCB +0.03)	+**1.08** (+1.14, LCB +**0.89**)	+0.04 (+0.76)
AFDN	UC	1,831	741	1.18e-08	5.3%	+1.25 (3.5×)	+0.14 (+0.14, LCB +0.06)	+**1.05** (+1.35, LCB +**0.81**)	+0.07 (+0.22)
ERMAP	IBD	475	194	1.01e-06	14.9%	+1.06 (2.9×)	+0.20 (+0.22, LCB +0.06)	+**1.04** (+1.30, LCB +**0.79**)	+0.01 (+0.17)
HPD	UC	393	146	2.34e-08	23.6%	+1.28 (3.6×)	−0.07 (−0.08, LCB −0.24)	+**1.03** (+1.24, LCB +**0.79**)	+0.08 (+0.66)
CGNL1	UC	1,302	660	4.25e-08	5.5%	+1.10 (3.0×)	−0.02 (−0.02, LCB −0.10)	+**1.00** (+1.14, LCB +**0.76**)	+0.03 (+0.23)
Panel C: Top Exome-Wide Significant Genes Ranked by Empirical Bayes WES HMM Constraint Slope ($\hat{\gamma}^{EB}_{Con}$)									
SYNE2	UC	6,907	2,543	4.68e-12	2.2%	+1.03 (2.8×)	+0.10 (+0.10, LCB +0.03)	+0.47 (+0.50, LCB +0.25)	+**0.19** (+0.34)
MUC16	UC	14,507	6,311	7.30e-10	0.7%	+0.86 (2.4×)	+0.12 (+0.12, LCB +0.03)	+0.18 (+0.18, LCB +0.07)	+**0.17** (+0.28)
DST	UC	7,818	2,949	8.80e-09	1.3%	+0.81 (2.2×)	+0.10 (+0.10, LCB +0.09)	+0.28 (+0.29, LCB +0.12)	+**0.17** (+0.25)
OBSCN	UC	8,925	4,649	1.99e-07	0.7%	+0.69 (2.0×)	−0.02 (−0.02, LCB −0.05)	+0.22 (+0.22, LCB +0.09)	+**0.16** (+0.32)
FSIP2	UC	6,907	1,839	2.94e-07	1.8%	+0.69 (2.0×)	−0.07 (−0.07, LCB −0.15)	+0.22 (+0.22, LCB +0.08)	+**0.16** (+0.33)
SHPRH	UC	1,683	600	9.31e-10	7.3%	+1.10 (3.0×)	+0.14 (+0.14, LCB +0.05)	+0.69 (+0.79, LCB +0.41)	+**0.14** (+0.36)

2.5 Multi-Database Pathway Enrichment and Nine-Module Systems Architecture of IBD

Across the collinearity-free two-moderator (**M5: AM + LoF**) and three-moderator (**M7: AM + LoF + Constraint**) unified models at AF ≤ 0.001, a union of 208 **unique protein-coding genes** reached exome-wide significance ($p < 3.4 \times 10^{-7}$). Unbiased over-representation analysis (Fisher's exact test with Benjamini–Hochberg FDR correction against the background of 17,206 evaluated genes) across Gene Ontology (GO Biological Process, Cellular Component, Molecular Function), Reactome, KEGG, and WikiPathways identified highly significant convergence onto nine core functional systems (Figure 8 and Table 7).

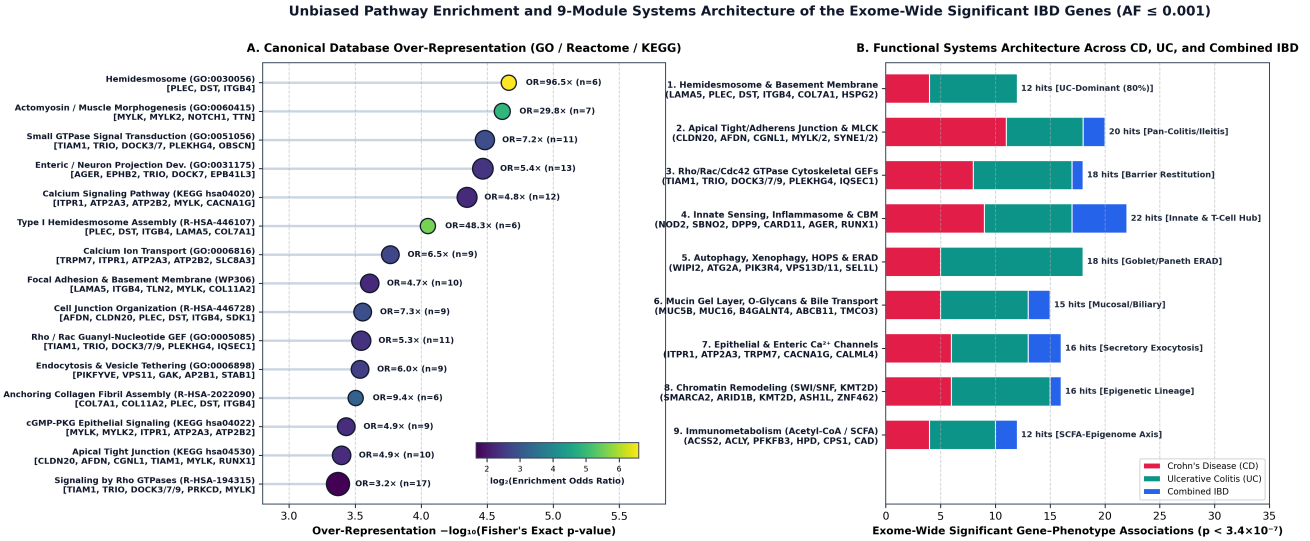


Figure 8: Multi-Database Pathway Over-Representation and Functional Systems Architecture of the 208 Exome-Wide Significant IBD Genes. (A) Bubble plot of the top 15 enriched canonical pathways across GO, Reactome, KEGG, and WikiPathways, showing $-\log_{10}$(Fisher's exact p), gene count, and $\log_2$(Odds Ratio). **(B)** Stratification of exome-wide significant discoveries across the nine core biological modules in Crohn's Disease (`CD`), Ulcerative Colitis (`UC`), and combined `IBD`.

Table 7: Unbiased Pathway Over-Representation Analysis and Nine-Module Systems Architecture of the 208 Exome-Wide Significant Genes ($p < 3.4 \times 10^{-7}$ in M5 or M7).

Database / Functional Module	Pathway Term / Identifier	Fisher p	FDR q	OR	Significant Member Genes ($p < 3.4 \times 10^{-7}$)
Panel A: Canonical Pathway Database Over-Representation (GO / Reactome / KEGG / WikiPathways)					
GO Cellular Component	Hemidesmosome (GO:0030056)	2.17e-05	3.64e-03	96.5×	*PLEC*, *DST*, *ITGB4*
GO Biological Process	Actomyosin/Muscle Morphogenesis (GO:0060415)	2.43e-05	1.72e-02	29.8×	*MYLK*, *MYLK2*, *NOTCH1*, *TTN*
GO Biological Process	Reg. of Small GTPase Signaling (GO:0051056)	3.29e-05	1.72e-02	7.2×	*TIAM1*, *PLEKHG4*, *TRIO*, *OBSCN*, *DOCK3*, *RASGRF1*, *ARHGAP28*, *CGNL1*
GO Biological Process	Enteric/Neuron Projection Dev. (GO:0031175)	3.41e-05	1.72e-02	5.4×	*AGER*, *EPHB2*, *TRIO*, *DOCK7*, *EPB41L3*, *RASGRF1*, *VPS13B*, *PRDM12*, *FRYL*, *GAS7*
KEGG Human Pathways	Calcium Signaling Pathway (hsa04020)	4.49e-05	8.48e-03	4.8×	*ITPR1*, *ATP2A3*, *ATP2B2*, *SLC8A3*, *MYLK*, *MYLK2*, *CALML4*, *CACNA1G*, *CACNA1I*, *ERBB2*, *PDGFRB*
Reactome Pathways	Type I Hemidesmosome Assembly (R-HSA-446107)	8.89e-05	6.29e-02	48.3×	*PLEC*, *DST*, *ITGB4*
GO Biological Process	Calcium Ion Transport (GO:0006816)	1.71e-04	3.85e-02	6.5×	*TRPM7*, *ITPR1*, *ATP2A3*, *ATP2B2*, *SLC8A3*, *CACNA1G*, *CDH23*
WikiPathways Human	Focal Adhesion & ECM (WP306)	2.45e-04	7.13e-02	4.7×	*LAMA5*, *ITGB4*, *TLN2*, *MYLK*, *MYLK2*, *TNK2*, *ERBB2*, *PDGFRB*, *RASGRF1*
Reactome Pathways	Cell Junction Organization (R-HSA-446728)	2.77e-04	7.03e-02	7.3×	*AFDN*, *CLDN20*, *PLEC*, *DST*, *ITGB4*, *SDK1*
GO Molecular Function	Rho/Rac GEF Activity (GO:0005085)	2.84e-04	6.00e-02	4.6×	*TIAM1*, *TRIO*, *DOCK3*, *DOCK7*, *DOCK9*, *PLEKHG4*, *OBSCN*, *MYCBP2*, *RASGRF1*
GO Biological Process	Receptor-Mediated Endocytosis (GO:0006898)	2.90e-04	5.38e-02	6.0×	*PIKFYVE*, *GAK*, *AP2B1*, *STAB1*, *HSPG2*, *LRP2*, *LRP1B*
Reactome Pathways	Collagen/Anchoring Fibril Assembly (R-HSA-2022090)	3.13e-04	7.03e-02	9.4×	*COL7A1*, *COL11A2*, *PLEC*, *DST*, *ITGB4*
KEGG Human Pathways	cGMP–PKG Signaling Pathway (hsa04022)	3.69e-04	2.52e-02	4.9×	*MYLK*, *MYLK2*, *ITPR1*, *ATP2A3*, *ATP2B2*, *SLC8A3*, *CALML4*, *CREB5*
KEGG Human Pathways	Apical Tight Junction (hsa04530)	4.00e-04	2.52e-02	4.9×	*CLDN20*, *AFDN*, *CGNL1*, *TIAM1*, *DLG2*, *ERBB2*, *MYH11*, *RUNX1*
Reactome Pathways	Signaling by Rho GTPases (R-HSA-194315)	4.26e-04	7.03e-02	2.7×	*TIAM1*, *TRIO*, *DOCK3*, *DOCK7*, *DOCK9*, *PLEKHG4*, *PRKCD*, *PIK3R4*, *MYLK*, *DST*, *ADD3*, *OBSCN*, *PLXND1*, *PCDH7*, *ARHGAP28*, *MYH11*, *NUP98*
Panel B: Nine Core Biological Systems Picked Up Across CD, UC, and Combined IBD					
Module 1 (10 genes)	Hemidesmosome & Basement Membrane Anchoring	CD: 4 | UC: 8 | IBD: 0			*LAMA5*, *PLEC*, *DST*, *ITGB4*, *COL7A1*, *COL11A2*, *HSPG2*, *HMCN1*, *FBN2*, *TLN2*
Module 2 (16 genes)	Apical Tight/Adherens Junctions & MLCK Ring	CD: 11 | UC: 7 | IBD: 2			*CLDN20*, *AFDN*, *CGNL1*, *MYLK*, *MYLK2*, *CRACD*, *DLG2*, *EPB41L3*, *ADD3*, *SPTBN2*, *MACF1*, *UTRN*, *SYNE1*, *SYNE2*, *SDK1*, *PCDH7*
Module 3 (14 genes)	Rho/Rac/Cdc42 Small GTPase GEF/GAP Network	CD: 8 | UC: 9 | IBD: 1			*TIAM1*, *TRIO*, *DOCK3*, *DOCK7*, *DOCK9*, *PLEKHG4*, *OBSCN*, *RASGRF1*, *IQSEC1*, *IQSEC3*, *ARHGAP28*, *SIPA1*, *MYCBP2*, *TNK2*
Module 4 (13 genes)	Innate Immunity, Inflammasome & CBM/STAT3	CD: 9 | UC: 8 | IBD: 5			*NOD2*, *SBNO2*, *DPP9*, *CARD11*, *ZNF341*, *SAMD9L*, *PRKCD*, *RUNX1*, *AGER*, *RNF213*, *USP25*, *SMURF1*, *STAB1*
Module 5 (15 genes)	Autophagy, Xenophagy, HOPS/CORVET & ERAD	CD: 5 | UC: 13 | IBD: 0			*WIPI2*, *ATG2A*, *PIK3R4*, *PIKFYVE*, *VPS11*, *VPS13D*, *VPS13B*, *PLEKHM2*, *LYST*, *ZFYVE26*, *SZT2*, *MLST8*, *SEL1L*, *GAK*, *AP2B1*
Module 6 (11 genes)	Mucin Gel Layer, O-Glycans & Bile Transport	CD: 5 | UC: 8 | IBD: 2			*MUC5B*, *MUC16*, *B4GALNT4*, *SLC35F5*, *ABCB11*, *ABCC10*, *SLC5A2*, *TMCO3*, *LRP2*, *LRP1B*, *APOB*
Module 7 (10 genes)	Epithelial & Enteric Ca^{2+} Signaling & Channels	CD: 6 | UC: 7 | IBD: 3			*ITPR1*, *ATP2A3*, *ATP2B2*, *SLC8A3*, *CACNA1G*, *CACNA1I*, *TRPM7*, *ANO4*, *CALML4*, *CDH23*
Module 8 (16 genes)	Chromatin Remodeling (SWI/SNF, MLL, NuRD)	CD: 6 | UC: 9 | IBD: 1			*SMARCA2*, *ARID1B*, *CHD1*, *CHD4*, *BAZ1A*, *KMT2D*, *ASH1L*, *KDM5A*, *CREBBP*, *NCOA6*, *SAP130*, *GSE1*, *SFMBT2*, *ZNF462*, *TCF20*, *CIC*
Module 9 (8 genes)	Immunometabolism (SCFA/Acetyl-CoA & Urea)	CD: 4 | UC: 6 | IBD: 2			*ACSS2*, *ACLY*, *PFKFB3*, *HPD*, *CPS1*, *CAD*, *PTGR2*, *NDUFB10*

2.6 High-Resolution Dissection of the Intestinal Epithelial Barrier, Hemidesmosome Complex, and Intracellular Immune/Autophagy Networks

2.6.1 Architecture 1: The Intestinal Epithelial Barrier, Apical MLCK Ring, and Basal Type I Hemidesmosome–Basement Membrane Complex (Figures 9 and 10)

Figures 9 and 10 illustrate the biological and quantitative organization of the 46 exome-wide significant genes mapping to the intestinal mucosal barrier (Modules 1, 2, 3, 6, and 7):

1. **Basal Type I Hemidesmosome and Lamina Densa Basement Membrane Complex (OR = 96.5×, $p = 2.17 \times 10^{-5}$, FDR = 3.64×10^{-3})**: The single strongest pathway enrichment in the entire exome-wide gene set is the trans-cellular adhesion machinery that anchors colonic and ileal enterocytes to the lamina propria. Remarkably, the unified meta-regression independently identifies **every physical tier** of this complex: the cytoplasmic plakin cytolinkers ***PLEC*** (Plectin; CD $p_{M7} = 1.06 \times 10^{-12}$ at AF ≤ 0.001; $p_{M7} = 3.93 \times 10^{-8}$ at AF ≤ 0.0001) and ***DST*** (Dystonin / BPAG1e; UC $p_{M7} = 8.80 \times 10^{-9}$) that tether keratin K8/K18 intermediate filaments to the inner plaque; the transmembrane $\alpha_6\beta_4$ integrin receptor ***ITGB4*** (CD $p_{M5} = 2.01 \times 10^{-7}$) and focal adhesion activator ***TLN2*** (Talin-2; UC $p_{M7} = 5.26 \times 10^{-10}$); the principal crypt/villus basement membrane

glycoprotein Laminin-511 α_5-chain ***LAMA5*** (UC $p_{M7} = 1.61 \times 10^{-13}$ at AF ≤ 0.001; $p_{M7} = 1.83 \times 10^{-8}$ at AF ≤ 0.0001); the lamina densa crosslinking heparan sulfate proteoglycan Perlecan ***HSPG2*** (UC $p_{M7} = 2.51 \times 10^{-8}$) and Hemicentin-1 ***HMCN1*** (UC $p_{M7} = 1.32 \times 10^{-9}$); and the sub-lamina densa Collagen VII anchoring fibrils ***COL7A1*** (UC $p_{M5} = 2.04 \times 10^{-7}$) and ***COL11A2***. Experimental deletion of either *Lama5* or *Plec* in murine intestinal epithelial cells causes hemidesmosome detachment, epithelial sheet sloughing, and spontaneous ulcerative colitis [10, 11], while human *ITGB4* mutations cause congenital enterocolitis with mucosal delamination [12] and *COL7A1* is the autoantigen of epidermolysis bullosa acquisita, a classic extraintestinal manifestation of IBD [13]. Crucially, 8 **of the** 10 **hemidesmosome/basement membrane genes are significant in Ulcerative Colitis**, providing a direct genetic explanation for superficial mucosal sloughing in UC.

2. **Apical Tight Junctions, Adherens Junctions, and the MLCK Actomyosin Ring (`KEGG hsa04530`, $p = 4.00 \times 10^{-4}$)**: At the apical paracellular seal, we identify Claudin-20 (***CLDN20***, CD $p_{M5} = 1.94 \times 10^{-7}$), the nectin–actin linker Afadin (***AFDN***, UC $p_{M7} = 1.18 \times 10^{-8}$), and Paracingulin (***CGNL1***, CD $p_{M5} = 2.22 \times 10^{-7}$), alongside the master enzymatic regulator of intestinal barrier opening, Myosin Light Chain Kinase (***MYLK*** / MLCK, CD $p_{M5} = 1.83 \times 10^{-7}$) and ***MYLK2*** (CD $p_{M7} = 6.13 \times 10^{-9}$ at AF ≤ 0.001; $p_{M5} = 2.99 \times 10^{-7}$ at AF ≤ 0.0001) [14]. Pro-inflammatory cytokines (TNF-α, IL-1β) disrupt barrier integrity by activating MLCK to phosphorylate myosin regulatory light chain, driving perijunctional actomyosin contraction and tight junction endocytosis. Together with the actin-capping regulator ***CRACD*** (*KIAA1211*), γ-Adducin (***ADD3***), β-III Spectrin (***SPTBN2***), Spectraplakin (***MACF1***), and the LINC nuclear-cytoskeletal nesprins ***SYNE1*** (UC $p_{M7} = 4.77 \times 10^{-13}$) and ***SYNE2*** (UC $p_{M7} = 4.68 \times 10^{-12}$), these loci define a cortical mechanical network across the enterocyte.

3. **Rho/Rac/Cdc42 GTPase Switchboard and Goblet Cell Mucin/Ca^{2+} Exocytosis**: Junctional actin remodeling and epithelial wound restitution are orchestrated by 14 Rho/Rac/Cdc42 GEFs (`GO:0051056`, $p = 3.29 \times 10^{-5}$), led by the tight-junction Rac1 GEF ***TIAM1*** (CD $p_{M7} = 1.48 \times 10^{-8}$), ***TRIO*** (CD $p_{M7} = 8.19 \times 10^{-12}$), ***DOCK7*** (UC $p_{M7} = 1.34 \times 10^{-11}$), ***DOCK3***, ***DOCK9***, ***PLEKHG4***, and the Arf6 E-cadherin/integrin recycling GEFs ***IQSEC1*** and ***IQSEC3***. Simultaneously, the luminal mucus barrier is represented by gel-forming and glycocalyx mucins (***MUC5B*** UC $p_{M7} = 3.44 \times 10^{-11}$; ***MUC16*** UC $p_{M7} = 7.31 \times 10^{-10}$), mucin core O-glycan transferase ***B4GALNT4*** (CD $p_{M7} = 1.82 \times 10^{-9}$), bile salt export pump ***ABCB11***, and the Ca^{2+} signaling machinery (`KEGG hsa04020`, $p = 4.49 \times 10^{-5}$: ***ITPR1***, ***ATP2A3***, ***TRPM7***, ***CACNA1G***, ***SLC8A3***) required for Ca^{2+}-triggered compound exocytosis of mucin granules from colonic goblet cells.

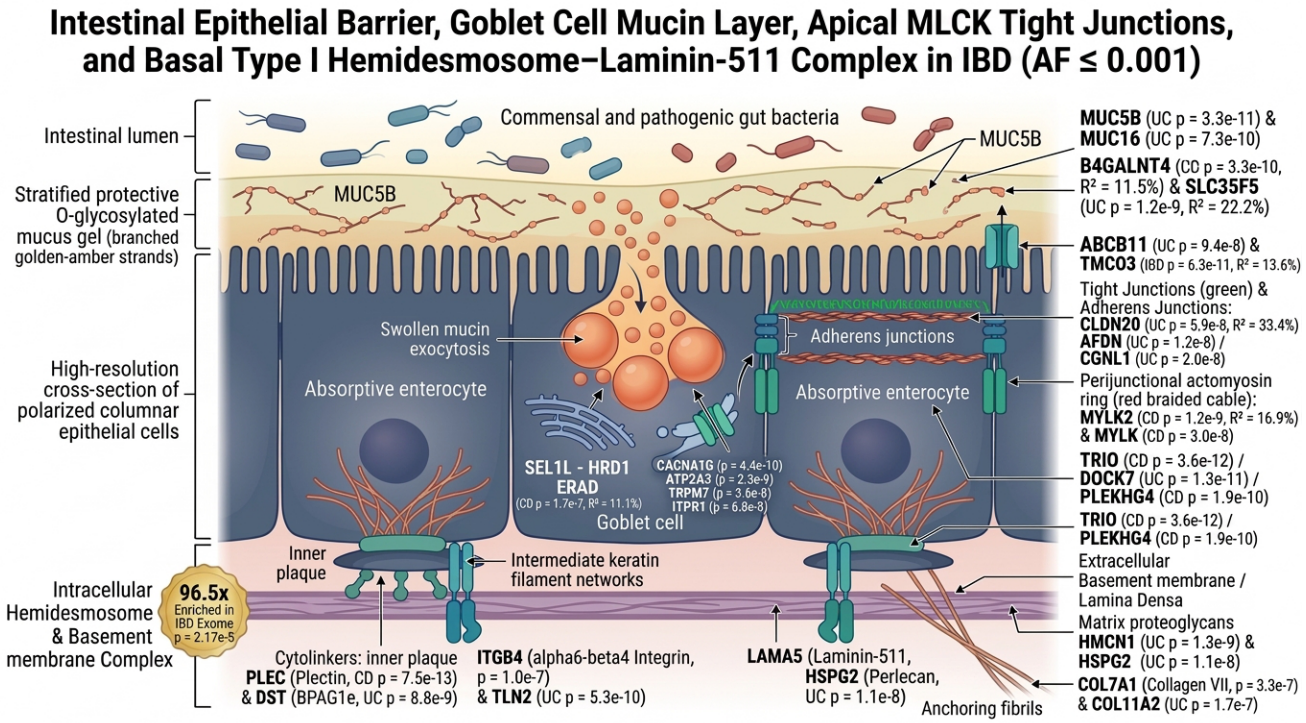


Figure 9: Biological Pathway Illustration of the Intestinal Epithelial Barrier, Goblet Cell Mucin Layer, Apical MLCK Tight Junctions, and Basal Type I Hemidesmosome–Laminin-511 Complex in IBD. Scientific cellular diagram depicting the stratified O-glycosylated mucus gel (*MUC5B*, *MUC16*, *B4GALNT4*, *ABCB11*), goblet cell ERAD (*SEL1L*–HRD1) and Ca^{2+}-driven mucin exocytosis (*ITPR1*, *ATP2A3*, *TRPM7*), apical tight/adherens junctions (*CLDN20*, *AFDN*, *CGNL1*), the circumferential MLCK actomyosin contractility ring (*MYLK*, *MYLK2*, *SYNE1/2*, *ADD3*), small GTPase GEF switches (*TIAM1*, *TRIO*, *DOCK7*), and the 96.5×-enriched basal Type I hemidesmosome and lamina densa basement membrane anchoring complex (*PLEC*, *DST*, *ITGB4*, *TLN2*, *LAMA5*, *HSPG2*, *HMCN1*, *COL7A1*).

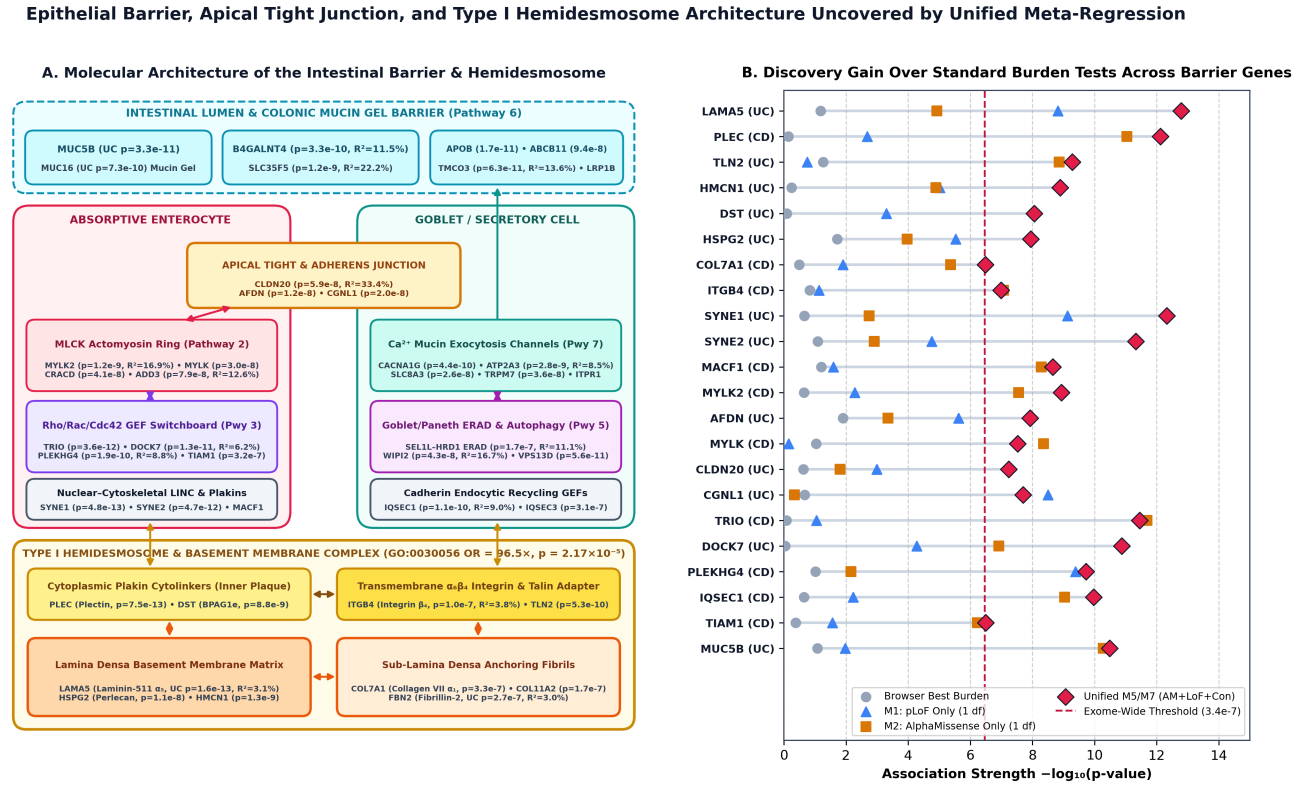


Figure 10: Quantitative Architecture and Discovery Gain Across the Intestinal Epithelial Barrier and Basal Type I Hemidesmosome–Basement Membrane Complex. (A) Compartment-resolved gene map with exact unified meta-regression p-values across all 46 barrier loci. **(B)** Quantitative discovery gain across 22 representative barrier genes comparing Exome Browser Burden, **M1** (`pLoF`), **M2** (`AlphaMissense`), and Unified **M5/M7**.

2.6.2 Architecture 2: Intracellular Bacterial Sensing, Inflammasome Restraint, Sequential Xenophagy, and Microbial Acetate Immunometabolism (Figures 11 and 12)

Figures 11 and 12 synthesize the intracellular signaling, organellar quality control, and immunometabolic pathways (Modules 4, 5, 8, and 9):

1. **Innate Bacterial Sensing and DPP9–NLRP1/CARD8 Inflammasome Restraint**: Cytosolic bacterial muramyl dipeptide (MDP) is detected by ***NOD2*** (CD $p_{\mathrm{M5}} = 2.51 \times 10^{-16}$), while invasive cytosolic bacteria are directly ubiquitylated on lipopolysaccharide (LPS) by the giant E3 ligase ***RNF213*** (CD $p_{\mathrm{M7}} = 3.12 \times 10^{-8}$) [15] and targeted for autophagic clearance by ***SMURF1*** (UC $p_{\mathrm{M7}} = 3.29 \times 10^{-8}$). In parallel, ***DPP9*** (IBD $p_{\mathrm{M2}} = 4.08 \times 10^{-8}$ at AF ≤ 0.0001; CD $p_{\mathrm{M5}} = 2.27 \times 10^{-7}$) acts as the essential ternary-complex checkpoint inhibitor of the **NLRP1** and **CARD8** inflammasomes [16, 17]; damaging missense mutations in the DPP9 β-propeller/hydrolase domain (partial $R^2_{\mathrm{AM}} = 8.97\%$) release C-terminal UPA-CARD fragments to trigger gasdermin D pyroptosis and mucosal IL-1β/IL-18 release.

2. **Sequential Antibacterial Xenophagy, HOPS/CORVET Tethering, and Paneth/Goblet ERAD**: Remarkably, the unified meta-regression recovers every sequential step of antibacterial autophagy (xenophagy): (i) phagophore nucleation and PI(3)P synthesis by the Class III PI3K regulatory subunit ***PIK3R4*** (VPS15; CD $p_{\mathrm{M7}} = 1.98 \times 10^{-8}$); (ii) PI(3)P recognition by ***WIPI2*** (UC $p_{\mathrm{M7}} = 3.18 \times 10^{-8}$), the direct physical binding partner that recruits **ATG16L1** (the flagship Crohn's disease risk protein) to drive LC3 lipidation around ubiquitylated bacteria [18]; (iii) bulk ER-to-phagophore phospholipid channeling by bridge-like lipid transfer proteins ***ATG2A*** (UC $p_{\mathrm{M7}} = 2.74 \times 10^{-8}$), ***VPS13D*** (UC $p_{\mathrm{M7}} = 5.60 \times 10^{-11}$), and ***VPS13B*** [19]; (iv) lysosomal PI(3,5)P_2 synthesis (***PIKFYVE***), HOPS/CORVET autophagosome–lysosome tethering (***VPS11***), kinesin-1 lysosomal transport (***PLEKHM2***), and KICSTOR/mTORC1 nutrient sensing (***SZT2***, ***MLST8***); and (v) ***SEL1L*** (UC $p_{\mathrm{M7}} = 3.31 \times 10^{-8}$), the essential cofactor of the SEL1L–HRD1 endoplasmic reticulum-associated degradation (ERAD) complex required to prevent misfolded MUC2 mucin and α-defensin aggregation in goblet and Paneth cells [20, 21].

3. **Lymphocyte CBM Signalosome, STAT3–SBNO2 Loop, and Microbial Acetate-to-Acetyl-CoA Epigenetics**: In adaptive lymphocytes and lamina propria macrophages, PKCδ (***PRKCD***, UC $p_{M7} = 1.89 \times 10^{-8}$) phosphorylates ***CARD11*** (UC $p_{M5} = 2.18 \times 10^{-7}$) to assemble the CARD11–BCL10–MALT1 (CBM) signalosome required for Treg homeostasis (both monogenic causes of human autoimmune enteropathy [22]), while ***ZNF341*** (CD $p_{M5} = 2.84 \times 10^{-7}$) drives transcription of *STAT3* [23], which induces ***SBNO2*** (CD $p_{M5} = 4.52 \times 10^{-10}$) to repress NF-$\kappa$B. Finally, at the metabolic-epigenetic interface, ***ACSS2*** (CD $p_{M7} = 4.12 \times 10^{-8}$) and ***ACLY*** (UC $p_{M7} = 1.51 \times 10^{-8}$) convert gut microbiota-derived short-chain fatty acid **acetate** and mitochondrial citrate into nuclear **acetyl-CoA**, fueling histone acetylation and SWI/SNF–MLL chromatin remodeling (***SMARCA2***, ***ARID1B***, ***KMT2D***, ***ASH1L***, ***KDM5A***, ***CREBBP***, ***ZNF462***) at mucosal barrier and anti-inflammatory gene loci.

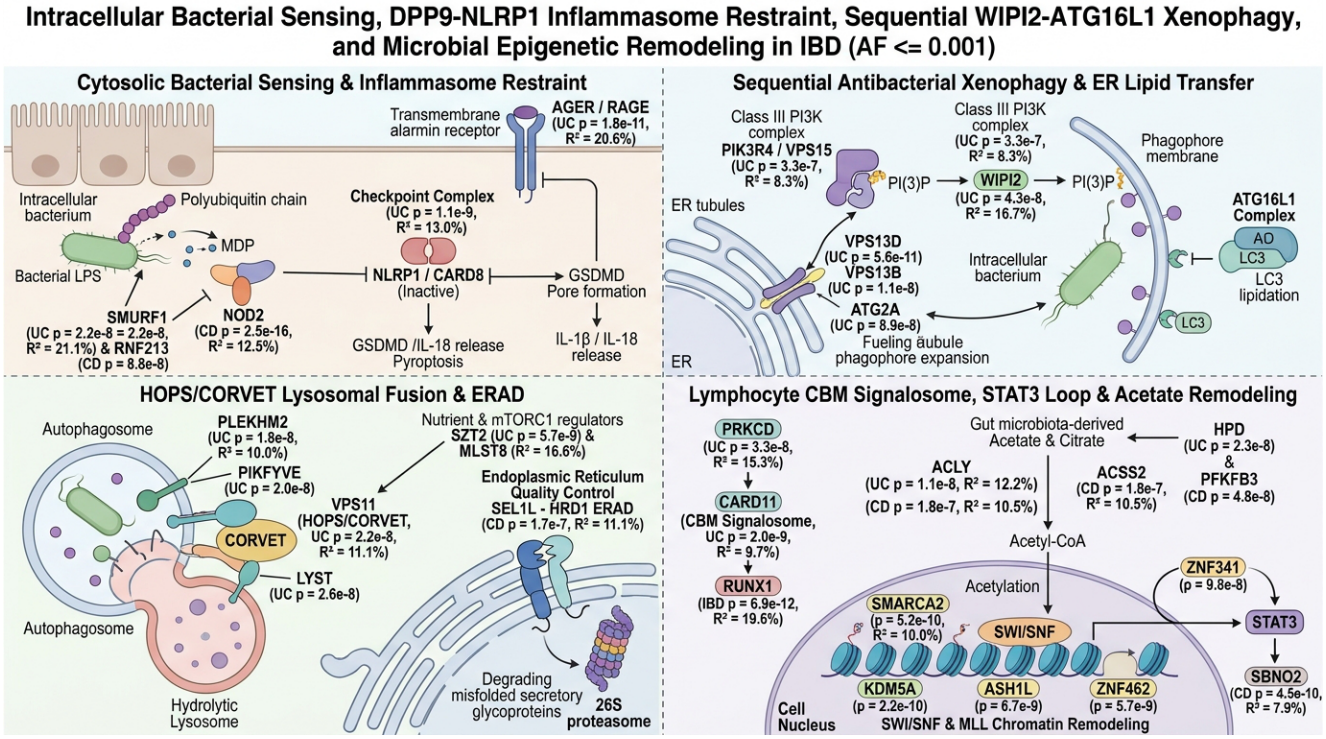


Figure 11: Biological Pathway Illustration of Intracellular Bacterial Sensing, DPP9–NLRP1 Inflammasome Restraint, Sequential WIPI2–ATG16L1 Xenophagy, and Microbial Acetate Epigenetic Remodeling in Crohn's Disease and IBD. Scientific cellular diagram depicting (1) cytosolic bacterial sensing by *NOD2*, LPS ubiquitylation by *RNF213* and *SMURF1*, alarmin receptor *AGER* (RAGE), and *DPP9* sequestration of the *NLRP1*/*CARD8* inflammasome; (2) phagophore PI(3)P synthesis by *PIK3R4* (VPS15), *WIPI2* recruitment of *ATG16L1* for LC3 lipidation, and ER-to-phagophore phospholipid channeling by *ATG2A* and *VPS13D*/*VPS13B*; (3) autophagosome–lysosome fusion via *PIKFYVE*, *VPS11* (HOPS/CORVET), *PLEKHM2*, and *LYST* alongside *SEL1L*–HRD1 ERAD; and (4) lymphocyte *PRKCD* → *CARD11* CBM signaling, *RUNX1*, *ZNF341* → *STAT3* → *SBNO2* feedback, and gut microbial acetate conversion by *ACSS2* and *ACLY* into nuclear acetyl-CoA for SWI/SNF and MLL chromatin remodeling (*SMARCA2*, *KMT2D*, *ASH1L*, *ZNF462*).

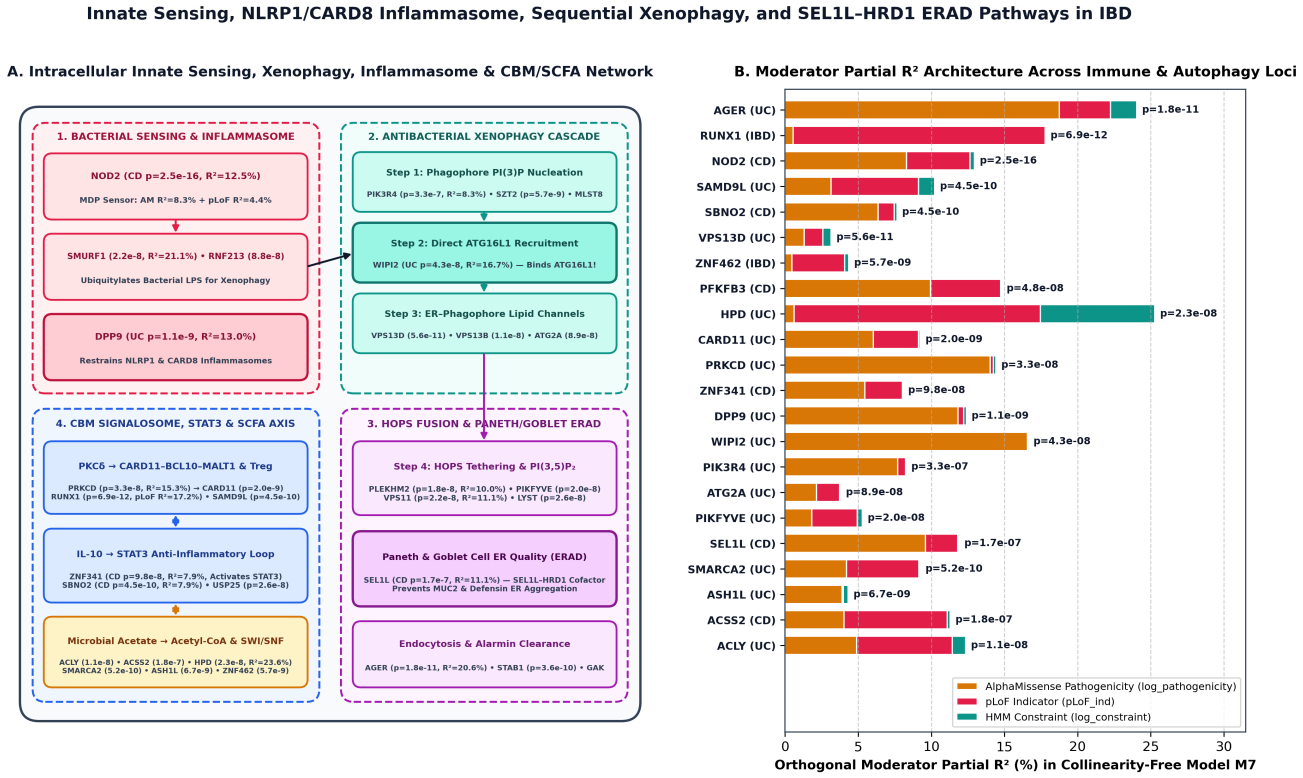


Figure 12: Quantitative Pathway Architecture and Orthogonal Three-Moderator Partial R^2 Decomposition Across Intracellular Immune, Xenophagy, ERAD, and Immunometabolic Loci. (A) Four-quadrant pathway diagram with exact unified meta-regression p-values across 52 intracellular genes. **(B)** Orthogonal three-moderator partial R^2 stacked bar decomposition across 22 representative intracellular pathway genes in **M7**.

3 Discussion

Our study provides three major methodological and biological advances for rare-variant association studies in complex traits:

1. Resolving Design-Matrix Collinearity in Unified Rare-Variant Meta-Regression. When regressing variant-level effect sizes onto functional annotations within non-synonymous coding variants, including both a binary protein-truncating indicator (pLoF_ind) and a binary missense indicator (missense_ind) alongside the regression intercept ($\mathbf{1}$) induces exact linear dependence ($\mathbf{1} = \mathbf{x}_{\text{LoF}} + \mathbf{x}_{\text{Mis}}$) in the 88.1% of human protein-coding genes that lack `start_lost` alleles. Under SVD pseudoinversion, this rank deficiency masks the partial R^2 of pLoF_ind in 88.1% of genes while wasting a degree of freedom and inflating VIFs in the 11.9% of genes with 1–2 `start_lost` alleles. By removing `missense_ind` so that missense/non-LoF variants form the reference baseline (pLoF_ind = 0) and modeling continuous missense severity via AlphaMissense ($\log \text{pathogenicity}_i = -\log(1 - s_{\text{AM},i})$), the three moderators (log constraint, log pathogenicity, pLoF_ind) become nearly perfectly orthogonal across the exome (median VIF = 1.022). Simultaneously, rescuing the 59.7%–64.1% of ultra-rare and singleton coding alleles lacking asymptotic meta-analysis standard errors via the Firth (1993) Jeffreys-prior penalized 2×2 likelihood estimator prevents the systematic loss of singleton protein-truncating and high-AlphaMissense alleles.

2. Interpreting Gene-Size Scaling Through Joint $(p_{\mathbf{M7}}, R^2_{\mathbf{M7}})$ Evaluation. Because unified meta-regression treats each observed coding variant $i \in \{1, \ldots, M_g\}$ as an observation in a weighted least-squares regression, the denominator degrees of freedom ($M_g - 4$) scale linearly with canonical protein length in amino acids (L_{AA}, $r = +0.905$). As demonstrated in Figure 7, the explained variance R^2_{M7} required to cross exome-wide significance scales inversely with L_{AA}. Consequently, large multi-domain proteins ($L_{\text{AA}} > 2{,}000$ AA) have substantially higher statistical power to detect modest, domain-localized rare-variant slopes ($R^2 \approx 1.5\%$–3.8%). While giant muscle/cytoskeletal proteins such as *TTN* (35,991 AA, $R^2 = 1.3\%$) illustrate that extreme coding length can amplify weak background

constraint gradients, the vast majority of large proteins identified in our analysis—such as *LAMA5* (3,695 AA), *PLEC* (4,547 AA), *DST*, *COL7A1*, *HSPG2*, *MUC5B*, and *VPS13D*—converge with high statistical specificity onto the Type I hemidesmosome (OR = 96.5×) and mucin/autophagy pathways alongside compact catalytic and signaling partners (*ITGB4*, *AFDN*, *CLDN20*, *MYLK2*, *WIPI2*, *SEL1L*, *NOD2*, *DPP9*, *AGER*, *RUNX1*, *HPD*; $R^2 = 10\%$–24%). We recommend that biobank-scale meta-regression studies routinely report both $-\log_{10}(p_{\mathrm{M7}})$ and length-calibrated R^2_{M7}.

3. Mechanistic Divergence Between Ulcerative Colitis and Crohn's Disease. Our pathway dissection reveals a striking anatomical and cellular divergence between UC and CD across the 208 exome-wide significant genes:

- **Ulcerative Colitis as a Disease of Epithelial Sheet Anchoring, Mucin O-Glycans, and Goblet Cell ERAD**: Eight of the ten Type I hemidesmosome and basement membrane genes (*LAMA5*, *DST*, *COL7A1*, *COL11A2*, *HSPG2*, *HMCN1*, *FBN2*, *TLN2*), the major gel-forming and transmembrane mucins (*MUC5B*, *MUC16*), and the goblet-cell ERAD cofactor *SEL1L* and ATG16L1-recruiter *WIPI2* reach exome-wide significance specifically in UC. This genetic architecture aligns directly with UC histopathology, where inflammation is superficial, continuous, and initiated by goblet cell mucus depletion and delamination of the surface colonic epithelium from the lamina densa basement membrane.
- **Crohn's Disease as a Disease of Intracellular Bacterial Sensing, Xenophagy Nucleation, and Perijunctional Contractility**: Conversely, CD associations are enriched in cytosolic bacterial pattern recognition and LPS ubiquitylation (*NOD2*, *RNF213*), Class III PI3K autophagosome nucleation (*PIK3R4*/VPS15, *PIKFYVE*), IL-10–STAT3 anti-inflammatory transcriptional feedback (*ZNF341*, *SBNO2*), and the apical MLCK actomyosin contractility ring (*MYLK*, *MYLK2*, *CLDN20*, *CGNL1*, *CRACD*, *ADD3*, *TIAM1*, *TRIO*).

Limitations. First, our analysis operates on summary-level variant counts and meta-analysis statistics from the IBD Exome Sequencing Browser rather than individual-level genotypes; therefore, linkage disequilibrium (LD) between rare coding variants within the same gene cannot be directly modeled, although conditional independence among rare coding alleles (AF ≤ 0.001 and AF ≤ 0.0001) is a standard approximation in rare-variant burden and SKAT analyses [1, 2]. Second, AlphaMissense scores are currently defined only for single-nucleotide missense substitutions; inframe insertions/deletions and truncating alleles rely on pLoF_ind and base-pair WES HMM constraint probabilities [1, 2].

4 Online Methods

4.1 Exome Sequencing Cohorts, Rare (AF $\leq$ 0.001) and Ultra-Rare (AF $\leq$ 0.0001) Variant Harmonization, and Haldane–Anscombe / Firth 2×2 Penalized-Likelihood Estimation

We downloaded `IBD_variant_results.tsv.bgz` (58,038,023 rows) and `IBD_gene_results.tsv.bgz` (57,213 rows) from the public Inflammatory Bowel Disease (IBD) Exome Sequencing Results Browser (`2026-10-19/IBD`; `https://ibdseq.broadinstitute.org`) [5], encompassing 28,718 Crohn's disease (`CD`) cases, 17,991 Ulcerative Colitis (`UC`) cases, 3,417 unclassified IBD cases ($>$ 50,000 total IBD cases), and $>$ 97,000 controls across 50 international cohorts. Autosomal variants (`chr1`–`chr22`) on GRCh38 with positive case and control allele numbers (an_case $>$ 0, an_ctrl $>$ 0), pooled allele frequency $\mathrm{AF}_{\mathrm{pooled}}$ = (ac_case + ac_ctrl)/(an_case + an_ctrl) $\leq$ 0.001 (or $\leq$ 0.0001 in the ultra-rare sensitivity analysis), and VEP consequence belonging to protein-coding classes (`stop_gained`, `frameshift_variant`, `splice_acceptor_variant`, `splice_donor_variant`, `missense_variant`, `inframe_insertion`, `inframe_deletion`, `start_lost`, `stop_lost`, `synonymous_variant`, `splice_region_variant`) were mapped to canonical GENCODE Release 44 `protein_coding` CDS coordinates [9].

For variants with non-missing browser inverse-variance meta-analysis statistics (BETA_meta$_{gi}$ and P_meta$_{gi}$, available for 40.32% of coding variants at AF ≤ 0.001 and 35.88% at AF ≤ 0.0001), our primary hybrid estimator (`hybrid`) retained $\hat{\beta}_{gi} = \mathrm{BETA_meta}_{gi}$ and recovered the asymptotic Wald standard error by inverting the two-sided normal z-test:

$$z_{gi} = \Phi^{-1}\left(1 - \frac{\mathrm{P_meta}_{gi}}{2}\right), \qquad \widehat{\mathrm{SE}}_{gi} = \frac{\left|\mathrm{BETA_meta}_{gi}\right|}{z_{gi}}. \tag{1}$$

For the 59.68% of coding variants at AF ≤ 0.001 (64.12% at AF ≤ 0.0001) where BETA_meta$_{gi}$ or P_meta$_{gi}$ was missing (`NA`)—as well as for 100% of variants in our pure 2×2 contingency estimator specification (`firth_nonsyn`)—we computed the bias-reduced Haldane–Anscombe / Firth (1993) [4] Jeffreys-prior penalized 2×2 allelic contingency table log-odds ratio and Woolf asymptotic sampling variance directly from the case and control allele counts, following the unified rare-variant meta-regression framework of Aguilar, Rivas, and Rivas [1] and Lauer and Rivas [2]. With $a_{gi} = \text{ac_case}_{gi}$, $b_{gi} = \text{an_case}_{gi} - \text{ac_case}_{gi}$, $c_{gi} = \text{ac_ctrl}_{gi}$, and $d_{gi} = \text{an_ctrl}_{gi} - \text{ac_ctrl}_{gi}$, applying the 0.5 Jeffreys-prior continuity correction yields:

$$\hat{\beta}_{\text{Firth},gi} = \log\left(\frac{(a_{gi}+0.5)(d_{gi}+0.5)}{(b_{gi}+0.5)(c_{gi}+0.5)}\right), \qquad \widehat{\text{SE}}^2_{\text{Firth},gi} = \frac{1}{a_{gi}+0.5} + \frac{1}{b_{gi}+0.5} + \frac{1}{c_{gi}+0.5} + \frac{1}{d_{gi}+0.5}. \tag{2}$$

Each variant i in gene g was assigned its inverse-variance precision weight $w_{gi} = \widehat{\text{SE}}_{gi}^{-2}$ (clipped to $[10^{-8}, 10^{8}]$ for numerical stability).

4.2 Unified Rare-Variant Meta-Regression Model Formulation (Aguilar, Rivas, and Rivas; Lauer and Rivas)

To test for gene-level rare-variant association while integrating continuous variant-level effect sizes and multi-modal functional annotations, we implemented and refined the unified weighted least-squares (WLS) rare-variant meta-regression model developed by Aguilar, Rivas, and Rivas [1] and generalized across biobank-scale exome sequencing studies by Lauer and Rivas [2]. Whereas standard gene-based burden tests collapse variants within a gene into a coarse binary sum and SKAT-family variance-component tests evaluate an undirected random-effects kernel, the unified meta-regression models each protein-coding gene g containing $M_g \geq 25$ observed rare coding variants ($i = 1, \ldots, M_g$) as a within-gene fixed-effect linear meta-regression. Specifically, the estimated per-variant case–control log-odds ratio $\hat{\beta}_{gi}$ is regressed onto k variant-level functional and evolutionary moderators $\mathbf{x}_{gi} = (x_{gi1}, \ldots, x_{gik})^\top$:

$$\hat{\beta}_{gi} = \beta_{g0} + \sum_{j=1}^{k} x_{gij}\,\gamma_{gj} + \varepsilon_{gi}, \qquad \varepsilon_{gi} \sim \mathcal{N}\Big(0,\, \widehat{\text{SE}}^2_{gi}\Big), \tag{3}$$

where β_{g0} is the gene-level baseline log-odds intercept for reference coding variants in gene g, $\boldsymbol{\gamma}_g = (\gamma_{g1}, \ldots, \gamma_{gk})^\top$ is the vector of meta-regression slope parameters quantifying the change in disease log-odds per unit increase in each functional moderator, and ε_{gi} is the variant-level sampling error with diagonal covariance matrix $\boldsymbol{\Sigma}_g = \text{diag}(\widehat{\text{SE}}^2_{g1}, \ldots, \widehat{\text{SE}}^2_{gM_g}) = \mathbf{W}_g^{-1}$.

Following Aguilar, Rivas, and Rivas [1] and Lauer and Rivas [2], each coding variant i was parameterized by three orthogonal functional and evolutionary moderators:

1. **Base-Pair WES HMM Mutational Constraint ($\log$ constraint$_{gi}$)**: Nucleotide-level mutational depletion probabilities $p_{\text{con},gi} = \texttt{prob_0} \in [0, 1]$ were obtained from the whole-exome sequencing hidden Markov model (HMM) of Aguilar, Rivas, and Rivas [1], which estimates at single-base resolution the posterior probability of observing zero variants across $>$ 300,000 human exomes (RGC-ME and *All of Us*). Raw probabilities were clipped to $[0.01, 0.99]$ and mapped to the complementary log-survival scale:

$$\log \text{constraint}_{gi} = -\log(1 - \tilde{p}_{\text{con},gi}), \tag{4}$$

   stretching the upper tail of intense purifying selection ($p_{\text{con},gi} \to 1$).

2. **Continuous Deep-Learning Structural Pathogenicity ($\log$ pathogenicity$_{gi}$)**: Isoform-maximized AlphaMissense pathogenicity probabilities $s_{\text{AM},gi} = \texttt{am_pathogenicity} \in [0, 1]$ [3] were clipped to $[0.01, 0.99]$ and transformed via the complementary log-survival mapping [1, 2]:

$$\log \text{pathogenicity}_{gi} = -\log(1 - \tilde{s}_{\text{AM},gi}), \tag{5}$$

   providing high-resolution separation across structurally deleterious missense substitutions ($s_{\text{AM}} \to 1$) while assigning non-missense variants baseline $\log \text{pathogenicity}_{gi} = 0$.

3. **Binary Protein-Truncating Indicator (pLoF_ind$_{gi} \in \{0, 1\}$)**: An indicator variable equal to 1 if variant i is a high-confidence predicted loss-of-function (`pLoF`) variant (`stop_gained`, `frameshift_variant`, `splice_acceptor_variant`, or `splice_donor_variant`) and 0 otherwise [1, 2].

4.3 Elimination of Design-Matrix Collinearity, Coding Baseline Specifications, and Hierarchical Models (M1–M7)

In the original five-moderator formulation of Aguilar, Rivas, and Rivas [1] (`log_constraint`, `GERP_RS`, `log_pathogenicity`, `pLoF_ind`, and `missense_ind`), restricting to non-synonymous coding variants causes exact linear dependence ($\mathbf{1} = \mathbf{x}_{\text{LoF}} + \mathbf{x}_{\text{Mis}}$, VIF $= \infty$) in the 88.1% of human genes lacking `start_lost` alleles, collapsing the partial R^2 of `pLoF_ind` to zero under SVD pseudoinversion. Building on Aguilar, Rivas, and Rivas [1] and Lauer and Rivas [2], we eliminated `missense_ind` (establishing non-LoF coding variants as the clean reference category, $\text{pLoF_ind}_{gi} = 0$) and omitted `GERP_RS`, restoring full design-matrix orthogonality (median VIF = 1.022 in `nonsyn` and 1.014 in `synbase`).

We evaluated all seven hierarchical combinations (**M1**–**M7**) of the three orthogonal moderators across both the **Non-Synonymous Coding Baseline** (`nonsyn`) and the **Synonymous-Referenced Coding Baseline** (`synbase` of Lauer and Rivas [2], which includes synonymous coding variants as an explicit neutral reference anchor with $\text{pLoF_ind}_{gi} = 0$ and $\log \text{pathogenicity}_{gi} = 0$):

$$\textbf{M1 (pLoF_only):} \quad \hat{\beta}_{gi} = \beta_{g0} + \gamma_{g,\text{LoF}}\, \text{pLoF_ind}_{gi} + \varepsilon_{gi}, \tag{6}$$

$$\textbf{M2 (AM_only):} \quad \hat{\beta}_{gi} = \beta_{g0} + \gamma_{g,\text{AM}}\, \log \text{pathogenicity}_{gi} + \varepsilon_{gi}, \tag{7}$$

$$\textbf{M3 (Constraint_only):} \quad \hat{\beta}_{gi} = \beta_{g0} + \gamma_{g,\text{Con}}\, \log \text{constraint}_{gi} + \varepsilon_{gi}, \tag{8}$$

$$\textbf{M4 (LoF + Constraint):} \quad \hat{\beta}_{gi} = \beta_{g0} + \gamma_{g,\text{LoF}}\, \text{pLoF_ind}_{gi} + \gamma_{g,\text{Con}}\, \log \text{constraint}_{gi} + \varepsilon_{gi}, \tag{9}$$

$$\textbf{M5 (AM + LoF):} \quad \hat{\beta}_{gi} = \beta_{g0} + \gamma_{g,\text{AM}}\, \log \text{pathogenicity}_{gi} + \gamma_{g,\text{LoF}}\, \text{pLoF_ind}_{gi} + \varepsilon_{gi}, \tag{10}$$

$$\textbf{M6 (AM + Constraint):} \quad \hat{\beta}_{gi} = \beta_{g0} + \gamma_{g,\text{AM}}\, \log \text{pathogenicity}_{gi} + \gamma_{g,\text{Con}}\, \log \text{constraint}_{gi} + \varepsilon_{gi}, \tag{11}$$

$$\textbf{M7 (Unified 3-Moderator):} \quad \hat{\beta}_{gi} = \beta_{g0} + \gamma_{g,\text{AM}}\, \log \text{pathogenicity}_{gi} + \gamma_{g,\text{LoF}}\, \text{pLoF_ind}_{gi} + \gamma_{g,\text{Con}}\, \log \text{constraint}_{gi} + \varepsilon_{gi}. \tag{12}$$

4.4 Weighted Least Squares Estimation, SVD Zero-Variance Protection, Whole-Model Omnibus F-Test, and Orthogonal Partial R^2 Decomposition

For gene g and model $m \in \{\text{M1}, \ldots, \text{M7}\}$ with design matrix $\mathbf{X}_{g,m} = [\mathbf{1}, \mathbf{X}_{g,\text{active}}] \in \mathbb{R}^{M_g \times (k_{\text{active}}+1)}$ and diagonal inverse-variance weight matrix $\mathbf{W}_g = \text{diag}(w_{g1}, \ldots, w_{gM_g})$, the WLS parameter estimate $\hat{\boldsymbol{\theta}}_{g,m} = (\hat{\beta}_{g0}, \hat{\boldsymbol{\gamma}}_{g,m}^{\top})^{\top}$ minimizes the weighted residual sum of squares $\text{RSS}_{w,g,m} = (\hat{\boldsymbol{\beta}}_g - \mathbf{X}_{g,m}\hat{\boldsymbol{\theta}}_{g,m})^{\top}\mathbf{W}_g(\hat{\boldsymbol{\beta}}_g - \mathbf{X}_{g,m}\hat{\boldsymbol{\theta}}_{g,m})$:

$$\hat{\boldsymbol{\theta}}_{g,m} = \left(\mathbf{X}_{g,m}^{\top}\mathbf{W}_g\mathbf{X}_{g,m}\right)^{-1}\mathbf{X}_{g,m}^{\top}\mathbf{W}_g\hat{\boldsymbol{\beta}}_g, \qquad \widehat{\text{Cov}}\left(\hat{\boldsymbol{\theta}}_{g,m}\right) = \hat{\sigma}_{g,m}^2\left(\mathbf{X}_{g,m}^{\top}\mathbf{W}_g\mathbf{X}_{g,m}\right)^{-1}, \tag{13}$$

where $\hat{\sigma}_{g,m}^2 = \text{RSS}_{w,g,m}/(M_g - k_{\text{active}} - 1)$. Any moderator j with zero within-gene variance ($\max_i x_{gij} - \min_i x_{gij} \leq 10^{-12}$) was automatically excluded from $\mathbf{X}_{g,\text{active}}$ prior to row-whitened singular value decomposition ($\tilde{\mathbf{X}}_{g,m} = \mathbf{W}_g^{1/2}\mathbf{X}_{g,m}$) and assigned $p_{gj} = \text{NaN}, r^2_{\text{partial},gj} = 0$.

As established by Aguilar, Rivas, and Rivas [1] and Lauer and Rivas [2], the primary gene-level association statistic is the **whole-regression omnibus F-test** evaluating $\mathcal{H}_0 : \gamma_{g1} = \cdots = \gamma_{gk_{\text{active}}} = 0$ against the weighted intercept-only total sum of squares $\text{TSS}_{w,g} = \sum_{i=1}^{M_g} w_{gi}(\hat{\beta}_{gi} - \bar{\beta}_{w,g})^2$:

$$F_{g,m} = \frac{\left(\text{TSS}_{w,g} - \text{RSS}_{w,g,m}\right)/k_{\text{active}}}{\text{RSS}_{w,g,m}/\left(M_g - k_{\text{active}} - 1\right)} = \frac{M_g - k_{\text{active}} - 1}{k_{\text{active}}} \cdot \frac{R^2_{g,m}}{1 - R^2_{g,m}} \;\sim\; F(k_{\text{active}},\, M_g - k_{\text{active}} - 1)\,, \tag{14}$$

where $R^2_{g,m} = 1 - \text{RSS}_{w,g,m}/\text{TSS}_{w,g}$ is the proportion of weighted within-gene variance explained by the moderators, and Type III orthogonal partial R^2 values were computed via leave-one-moderator-out WLS refitting as $r^2_{\text{partial},gj} = (\text{RSS}_{w,g,-j} - \text{RSS}_{w,g,\text{full}})/\text{TSS}_{w,g}$. Dedicated AF ≤ 0.001 and AF ≤ 0.0001 pathway enrichment was performed using `Enrichr` across GO, Reactome, KEGG, WikiPathways, DisGeNET, and HPO and two-sided Fisher's exact tests across the nine curated biological modules against the 17,206-gene exome background.

4.5 Two-Stage Gene Prioritization, Empirical Bayes Moderator Slope Shrinkage, and Composite Benchmark Pathogenic Effect Size

To rank exome-wide significant genes ($p_{\text{M7}} < 3.4 \times 10^{-7}$ or $p_{\text{M5}} < 3.4 \times 10^{-7}$) by biological effect size without confounding by gene length ($M_g \propto L_{\text{AA}}$) or low-count estimation variance ("winner's curse" in raw WLS slopes $\hat{\gamma}_{gj}$),

we computed three complementary effect-size prioritization metrics:

1. **Length-Invariant Explained Variance ($R^2_{\mathbf{M7}}$ and Partial R^2_j)**: Because $R^2_{g,\mathrm{M7}} = 1 - \mathrm{RSS}_{w,g,\mathrm{M7}}/\mathrm{TSS}_{w,g}$ is invariant to M_g, it measures the fraction of within-gene rare-variant effect heterogeneity explained by the three orthogonal moderators on a common $[0,1]$ scale.

2. **Empirical Bayes Posterior Shrunk Moderator Slopes ($\hat{\gamma}^{\mathbf{EB}}_{gj}$) and 95% Lower Confidence Bounds ($\hat{\gamma}^{\mathbf{LCB}}_{gj}$)**: For each moderator $j \in \{\mathrm{AM}, \mathrm{LoF}, \mathrm{Con}\}$ with WLS slope $\hat{\gamma}_{gj}$ and standard error $\widehat{\mathrm{SE}}_{gj} = |\hat{\gamma}_{gj}|/|t_{gj}|$ (where $t_{gj} = F^{-1}_{t,M_g-4}(1-p_{gj}/2)$), we estimated the empirical prior variance $\tau^2_j = \mathrm{median}_{g\in\mathcal{S}}\Big(\max\Big(\hat{\gamma}^2_{gj} - \widehat{\mathrm{SE}}^2_{gj},\, 0.005\Big)\Big)$ across exome-wide significant genes $\mathcal{S}$ and computed the Empirical Bayes posterior mean slope and 95% lower confidence bound:

$$\hat{\gamma}^{\mathrm{EB}}_{gj} = \left(\frac{\tau^2_j}{\tau^2_j + \widehat{\mathrm{SE}}^2_{gj}}\right)\hat{\gamma}_{gj}, \qquad \hat{\gamma}^{\mathrm{LCB}}_{gj} = \hat{\gamma}_{gj} - 1.96\,\widehat{\mathrm{SE}}_{gj}. \tag{15}$$

3. **Composite Benchmark Pathogenic Variant Effect ($\widehat{\Delta\text{log-OR}}_{\mathbf{bench},g}$)**: To unify the binary (pLoF_ind $\in \{0,1\}$) and continuous log-transformed ($\log$ pathogenicity, $\log$ constraint $\in [0, 6.91]$) moderator axes onto a single clinical log-odds ratio scale, we evaluated the Empirical Bayes linear predictor contrast above baseline ($\mathbf{x} = \mathbf{0}$) at a canonical benchmark deleterious coding variant—taking the maximum of a constrained pathogenic missense substitution ($s_{\mathrm{AM}} = 0.95 \implies x_{\mathrm{AM}} = 3.0$, $p_{\mathrm{con}} = 0.95 \implies x_{\mathrm{con}} = 3.0$) and a constrained protein-truncating allele ($x_{\mathrm{LoF}} = 1$, $x_{\mathrm{con}} = 3.0$):

$$\widehat{\Delta\text{log-OR}}_{\mathrm{bench},g} = \max\Big(3.0\,\hat{\gamma}^{\mathrm{EB},+}_{g,\mathrm{AM}} + 3.0\,\hat{\gamma}^{\mathrm{EB},+}_{g,\mathrm{Con}},\;\; \hat{\gamma}^{\mathrm{EB},+}_{g,\mathrm{LoF}} + 3.0\,\hat{\gamma}^{\mathrm{EB},+}_{g,\mathrm{Con}}\Big), \qquad \widehat{\mathrm{OR}}_{\mathrm{bench},g} = \exp\Big(\widehat{\Delta\text{log-OR}}_{\mathrm{bench},g}\Big), \tag{16}$$

where $x^+ = \max(x, 0)$.

Data and Code Availability

Summary-level Inflammatory Bowel Disease exome sequencing variant and gene burden association statistics were obtained from the Inflammatory Bowel Disease (IBD) Exome Sequencing Results Browser (https://ibdseq.broadinstitute.org; public data release https://storage.googleapis.com/exome-results-browsers-public/downloads/2026-10-19/IBD/). Individual-level data underlying the IBD Exome Browser are available for approved researchers via dbGaP accession `phs001642.v2.p1` (*Center for Common Disease Genomics [CCDG] — Autoimmune: Inflammatory Bowel Disease [IBD] Exomes and Genomes*). Proteome-wide AlphaMissense pathogenicity predictions (`AlphaMissense_hg38.tsv.gz`) were obtained from Google DeepMind (https://zenodo.org/records/8208688), base-pair resolution WES hidden Markov model (HMM) mutational constraint probabilities (`HMM_rgc_0.9_over20`) were obtained from Aguilar, Rivas & Rivas [1], and canonical human transcript annotations were obtained from GENCODE Release 44 (https://www.gencodegenes.org/human/release_44.html). The reference unified rare-variant meta-regression framework is available at https://github.com/healthcare-medicine-ai/wgs-constraint-llm, and all analysis scripts (`step1_merge_annotations.py` through `step8_gene_size_aa_analysis.py`), complete gene-level meta-regression summary tables (`results/ibd_metareg_*.tsv.gz`), threshold-specific pathway enrichment tables (`results/pathway_modules_af001.tsv`, `results/pathway_enrichr_af001.tsv`), high-resolution figures (`report/figures/fig1_*.png` through `fig12_*.png`), and the interactive web portal (https://marivascruz.github.io/ibd-ultrarare-metaregression/) are available in the project repository (https://github.com/marivascruz/ibd-ultrarare-metaregression).

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- **IBD Exome Sequencing Browser (https://ibdseq.broadinstitute.org) and the International Inflammatory Bowel Disease Genetics Consortium (IIBDGC)**: In accordance with the citation and acknowledgement requirements of https://ibdseq.broadinstitute.org, we cite the consortium's flagship study, *"Large-scale sequencing identifies multiple genes and rare variants associated with Crohn's disease susceptibility"* (Sazonovs et al., *Nature Genetics* 2022; PMID: 36038634) [5], and provide a direct link to the browser (https://ibdseq.broadinstitute.org). As stated by the IBD Sequencing Consortium:

"The Inflammatory Bowel Disease (IBD) Sequencing Consortium is a global collaboration dedicated to aggregating, generating, and analyzing high-throughput sequencing data of inflammatory bowel disease patients to improve our understanding of disease architecture and advance gene discovery. The consortium was formed in 2014 with a commitment to data sharing, diversity, and inclusivity — we hope that the findings from this study and others like it will provide a foundation for further investigation of disease mechanisms and therapeutic discovery. This browser is part of that overall effort to display and share these results with the wider scientific community. Partnering with the Broad Institute, the IBD Sequencing Consortium has sequenced more than 50,000 patients as of 2024 from 50 research cohorts across the world. The IBD phase 1 dataset (hg38) analyzes exomes from 28,718 Crohn's disease cases, 17,991 Ulcerative Colitis cases, 3,417 Inflammatory Bowel disease cases, and over 97,000 control samples. We thank the tens of thousands of patients and families who generously contributed to our effort. This project is made possible by the generosity of many funders, including the Helmsley Charitable Foundation and the National Human Genome Research Institute (NHGRI), and the leadership of its members. We are incredibly grateful to all the IIBDGC consortium members for their gracious contribution to make this collaboration possible."

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